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TAMASHIRO et al.(10) **Pub. No.: US 2025/0286278 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **ANTENNA DEVICE****Publication Classification**(71) Applicant: **TOKIN CORPORATION**,
Shiroishi-shi, Miyagi (JP)(51) **Int. Cl.**
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(JP); **Toshiaki OKA**, Sendai-shi (JP)(52) **U.S. Cl.**
CPC **H01Q 7/06** (2013.01)(73) Assignee: **TOKIN CORPORATION**,
Shiroishi-shi, Miyagi (JP)(57) **ABSTRACT**(21) Appl. No.: **18/858,402**(22) PCT Filed: **Apr. 27, 2023**(86) PCT No.: **PCT/JP2023/016764**

§ 371 (c)(1),

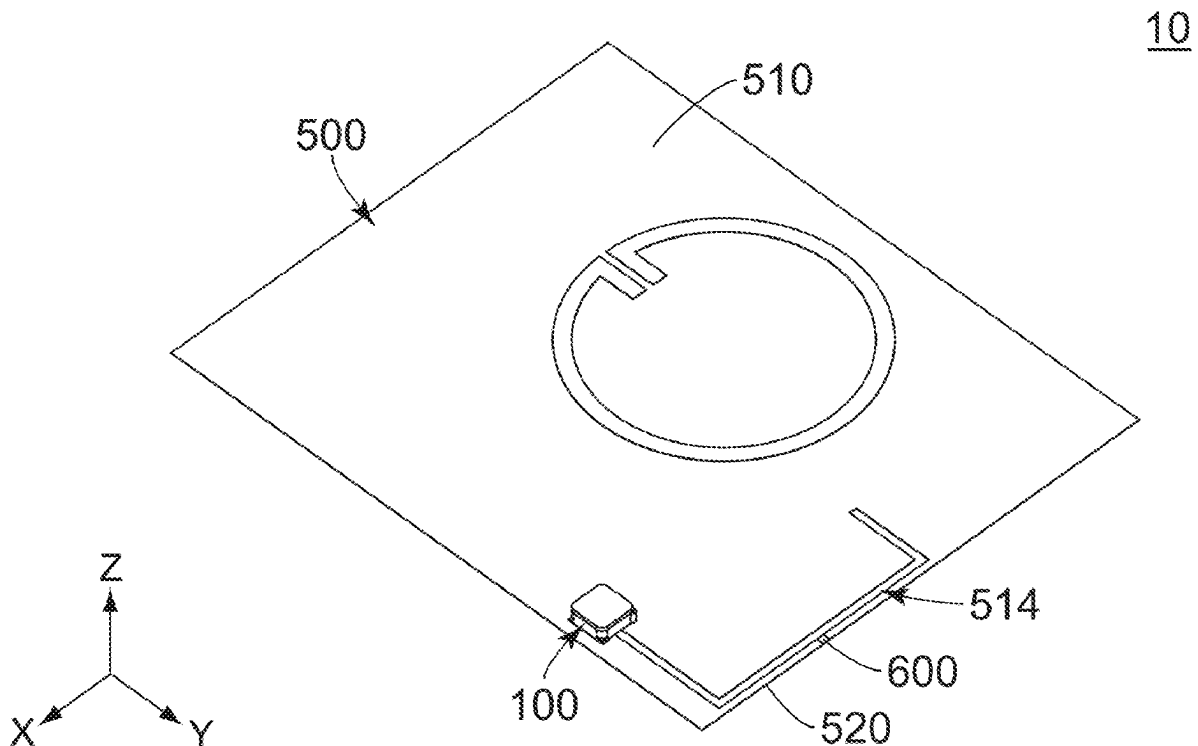
(2) Date: **Oct. 21, 2024**

An antenna device includes a circuit board, a coil antenna and a resonance circuit. The circuit board has a conductive portion with a flat shape. The conductive portion is formed with an opening and a slit. The slit is coupled with the opening. The coil antenna is mounted on the circuit board so that the coil antenna at least overlaps with the opening when the coil antenna is viewed along an up-down direction perpendicular to the circuit board. The coil antenna includes a coil and a magnetic core. The magnetic core partially forms a magnetic path of the coil. The magnetic core is positioned at a region excluding a specific region which is positioned just below the coil. The magnetic core at least has a center core. The center core is at least partially positioned inside the coil.

(30) **Foreign Application Priority Data**

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Jul. 26, 2022 (JP) 2022-118580



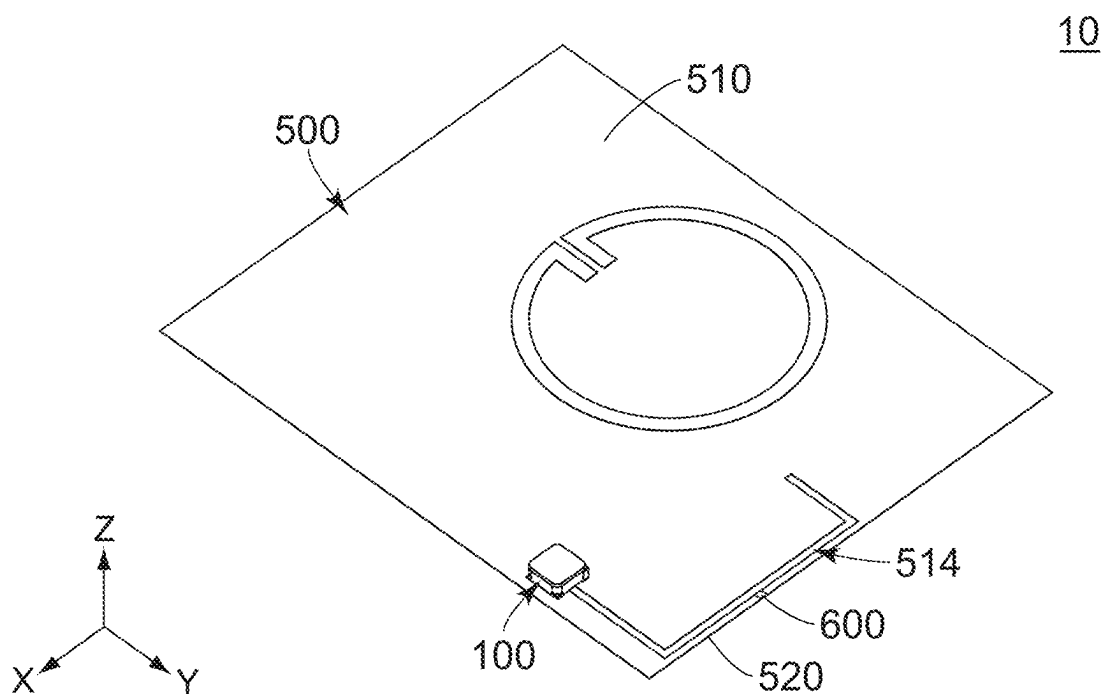


FIG. 1

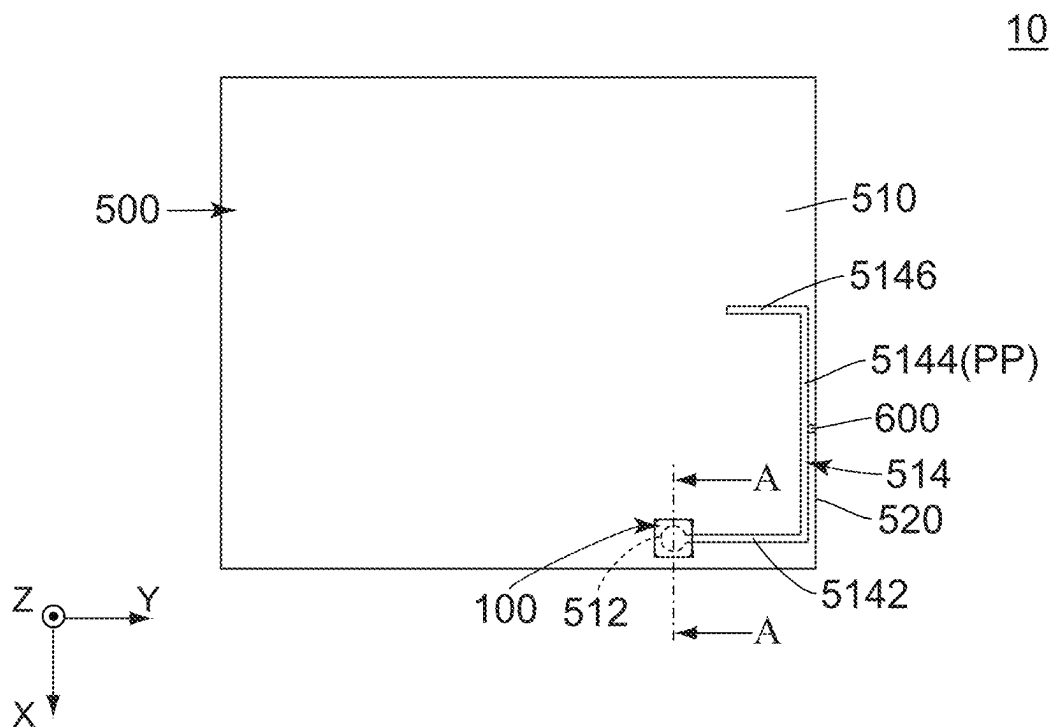


FIG. 2

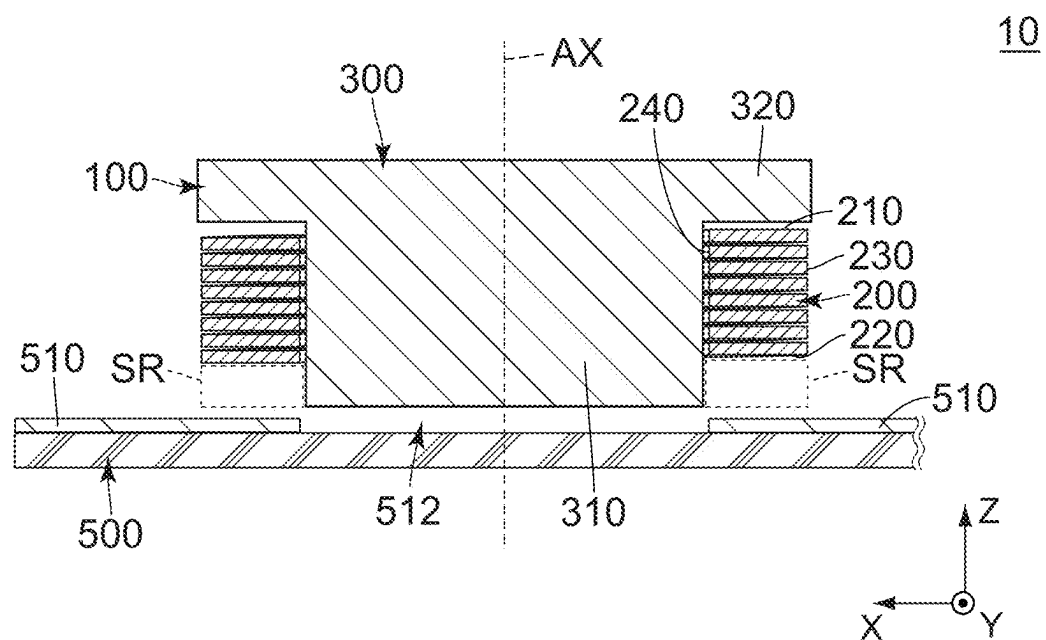


FIG. 3

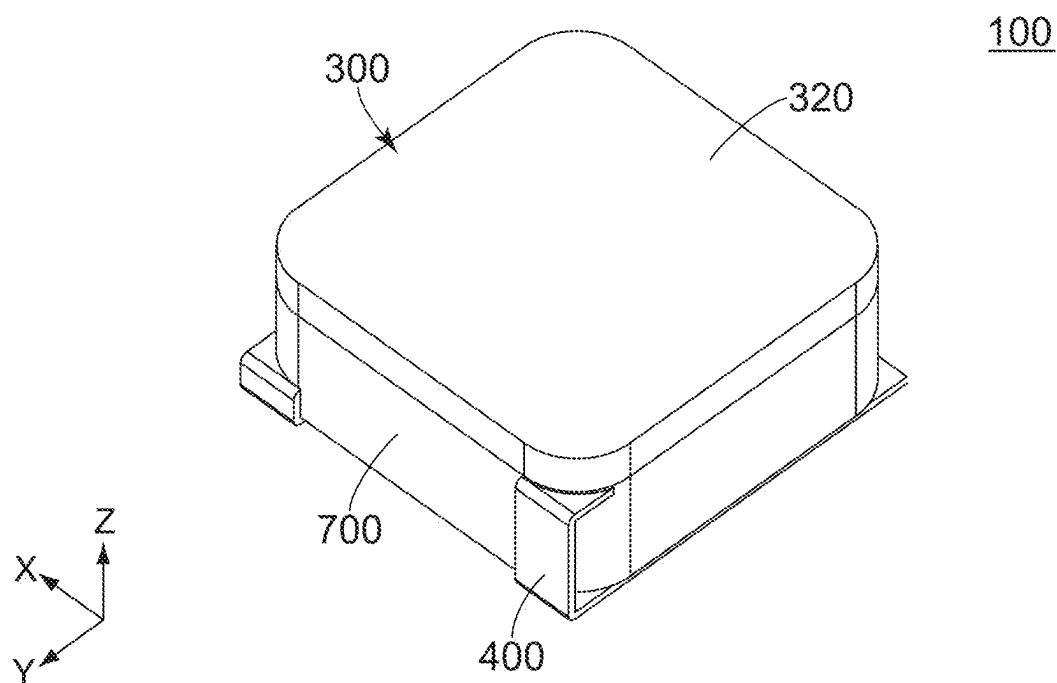


FIG. 4

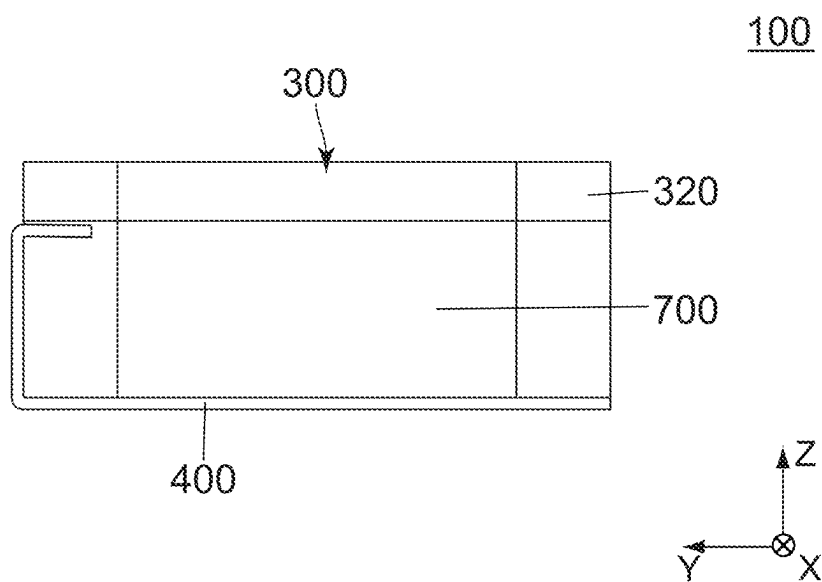


FIG. 5

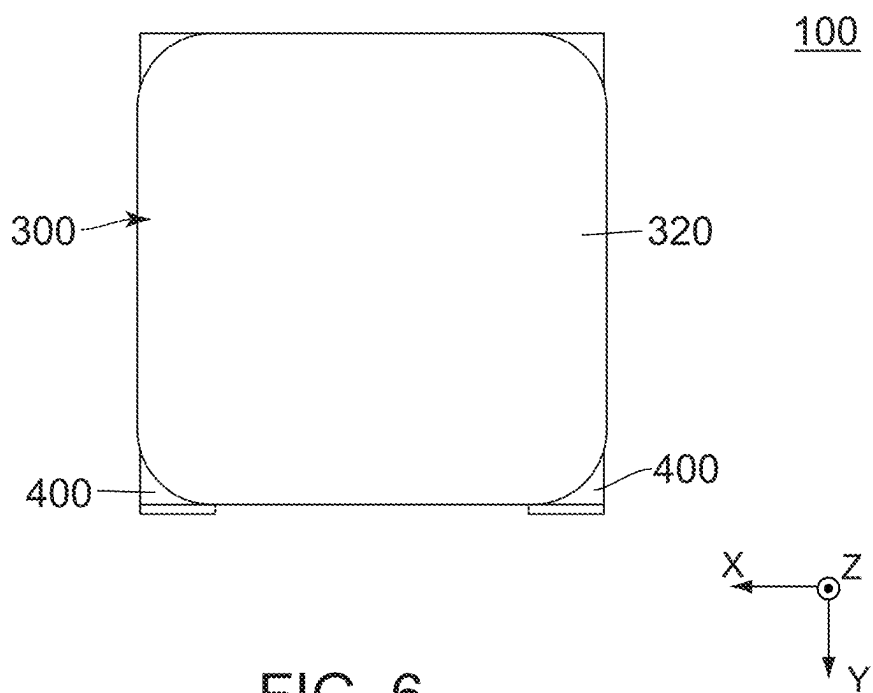


FIG. 6

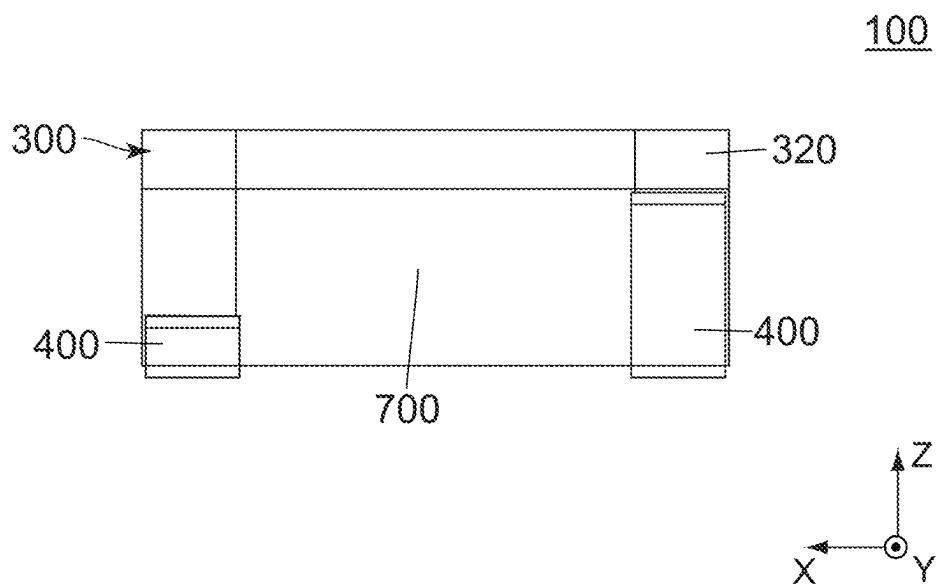


FIG. 7

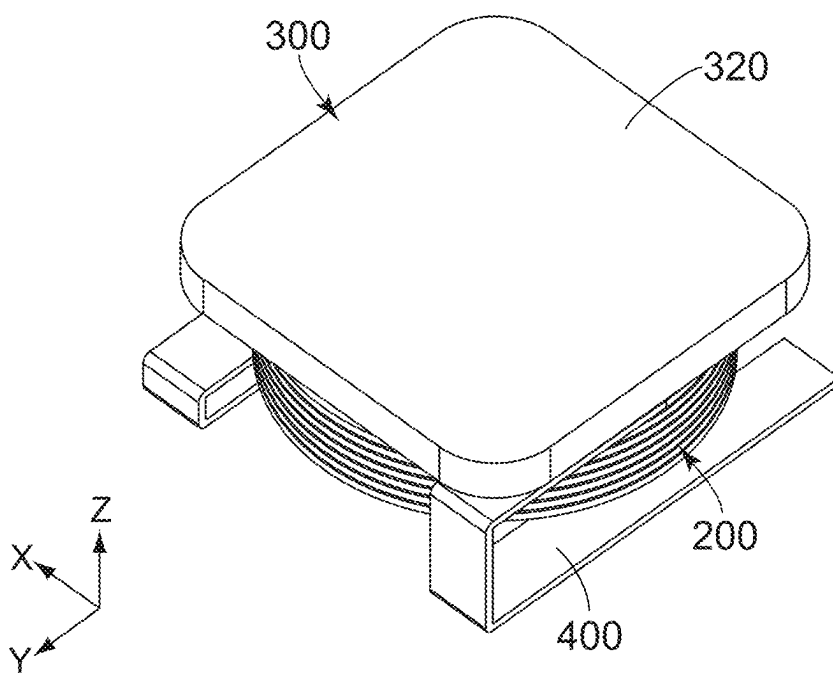


FIG. 8

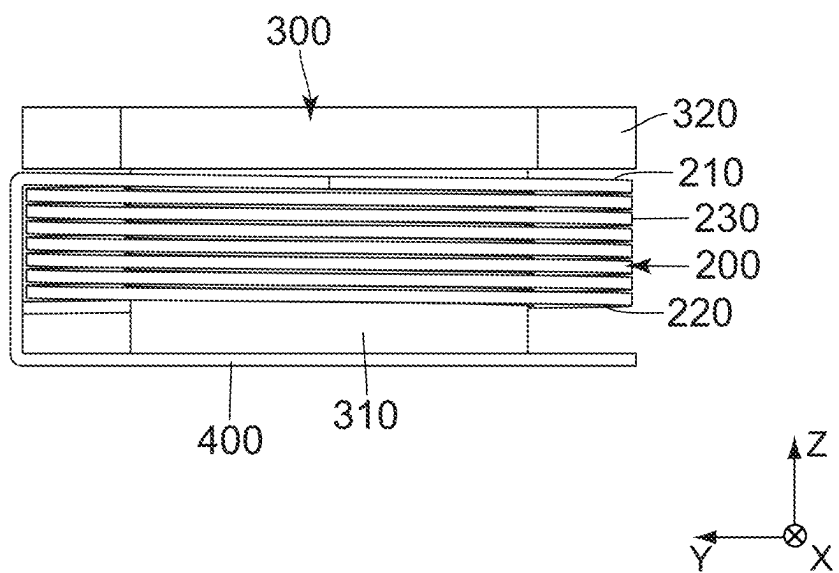


FIG. 9

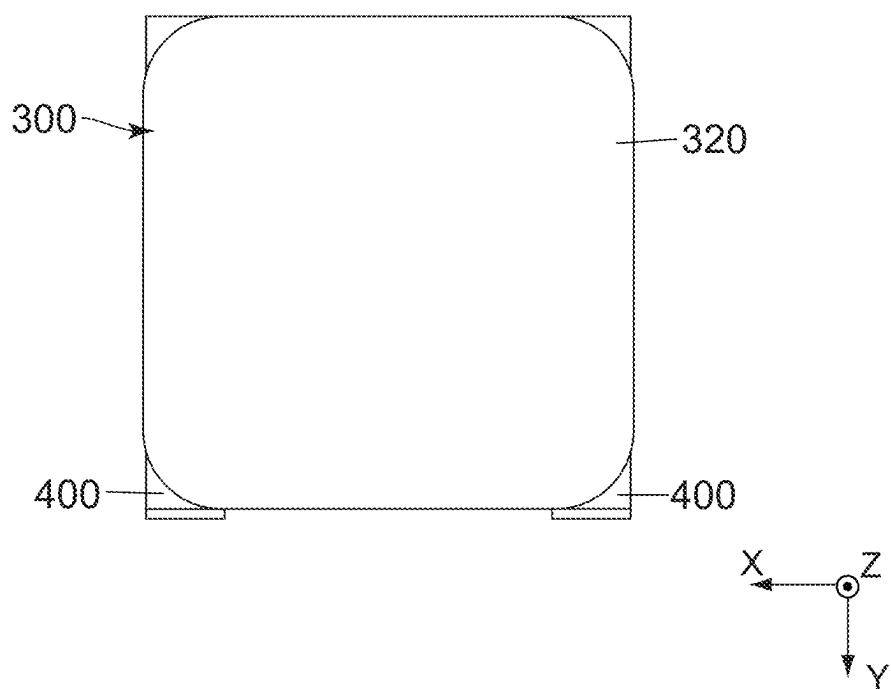


FIG. 10

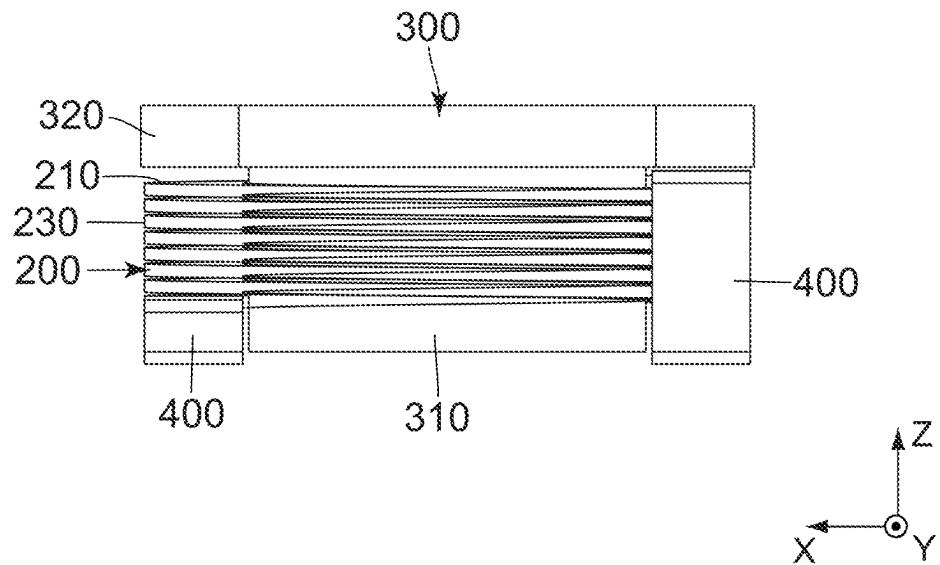


FIG. 11

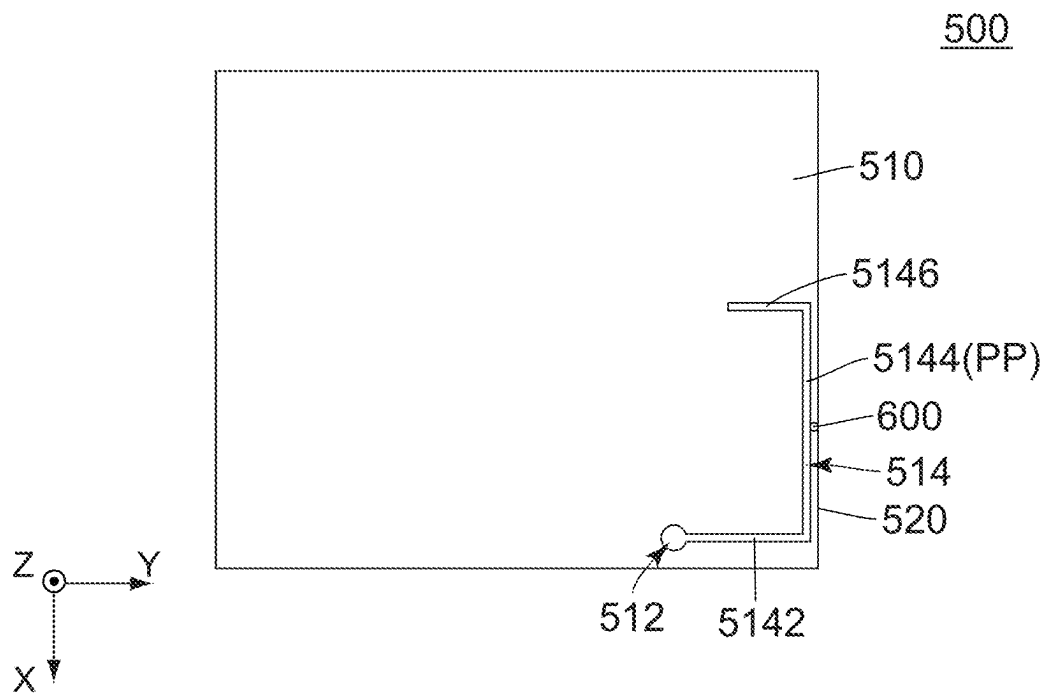


FIG. 12

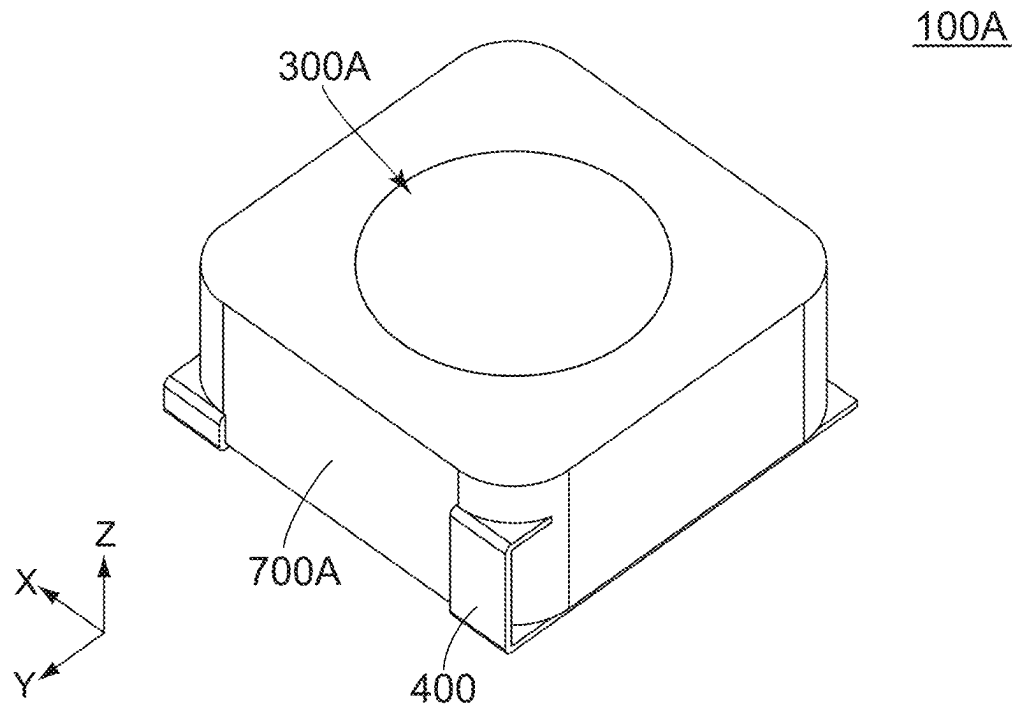


FIG. 13

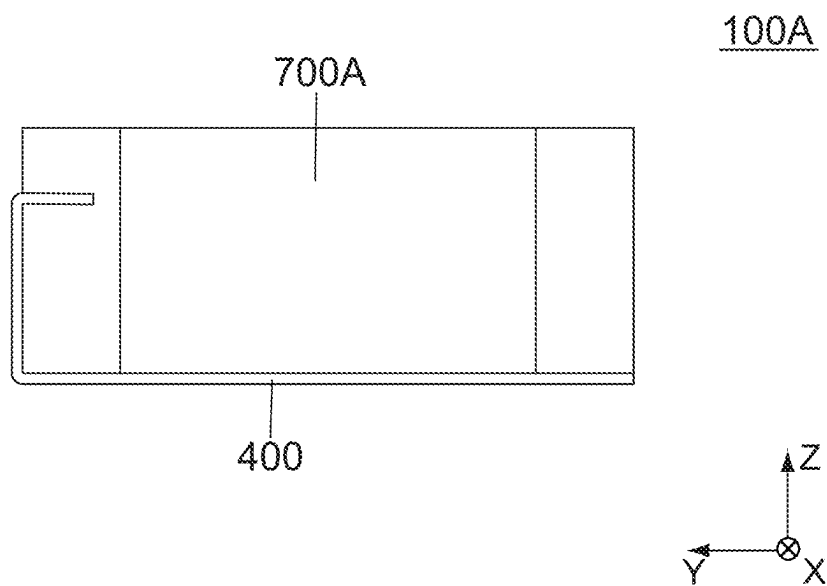


FIG. 14

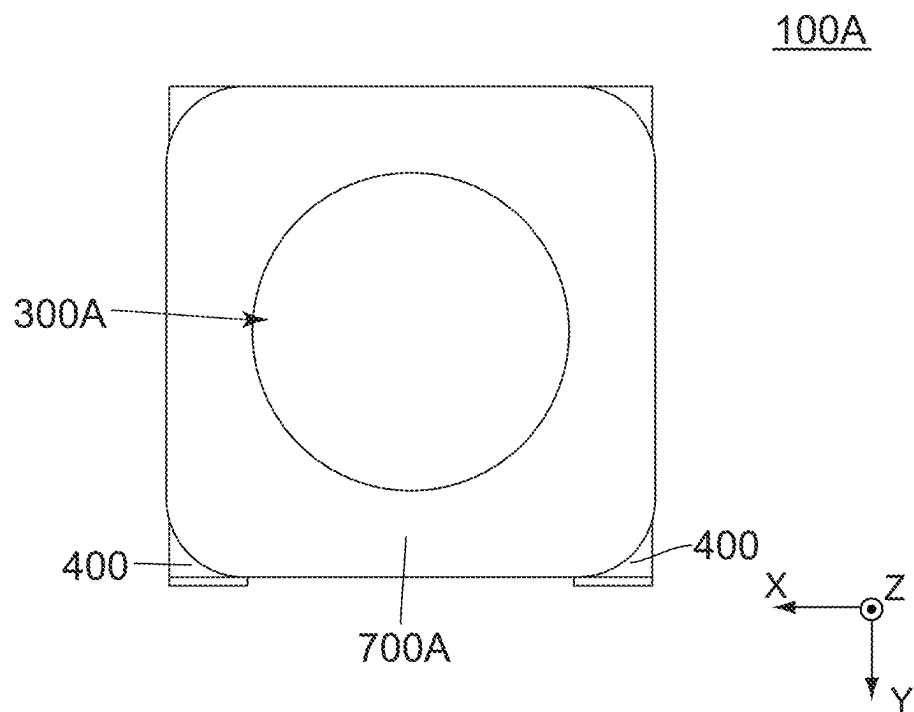


FIG. 15

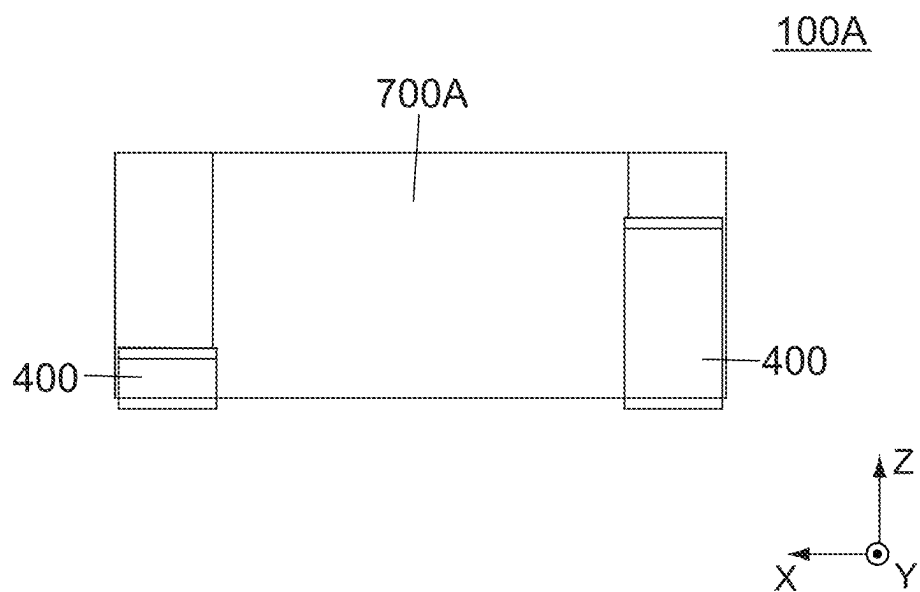


FIG. 16

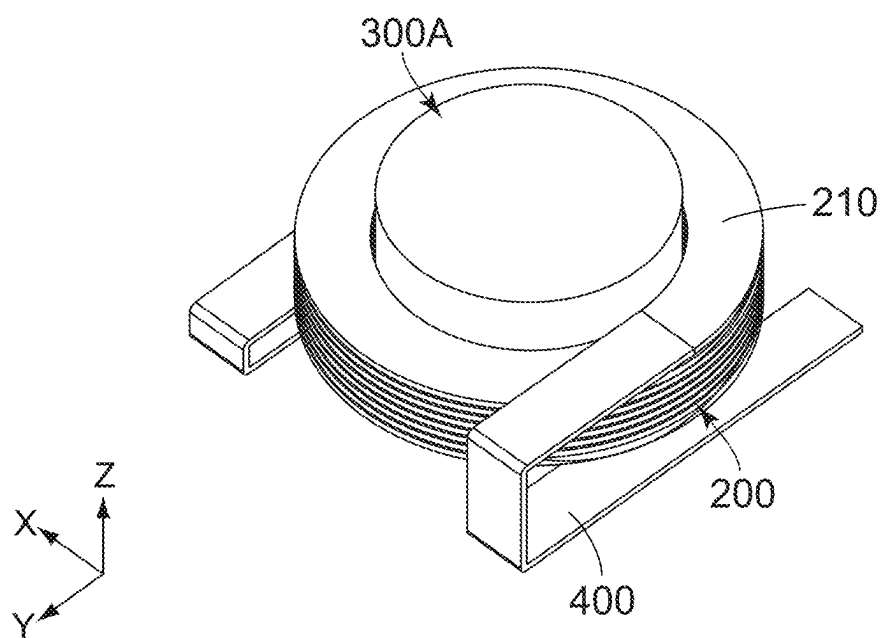


FIG. 17

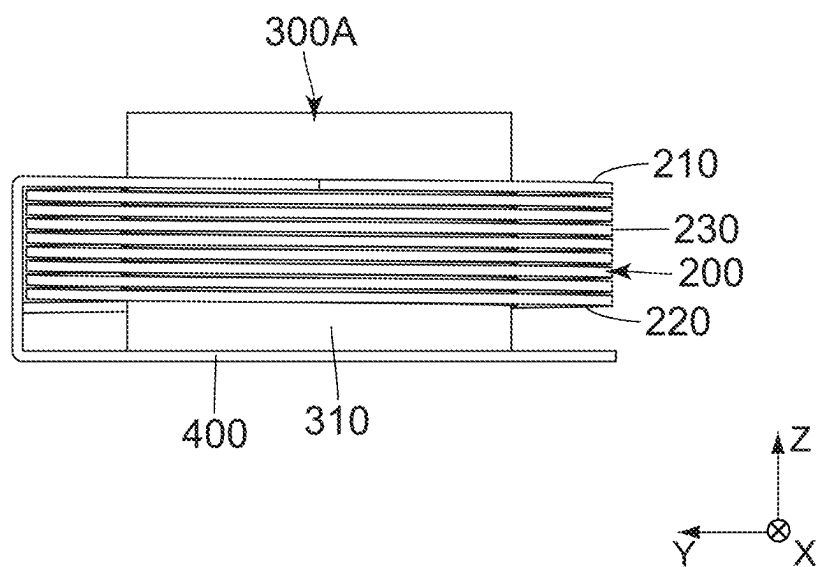


FIG. 18

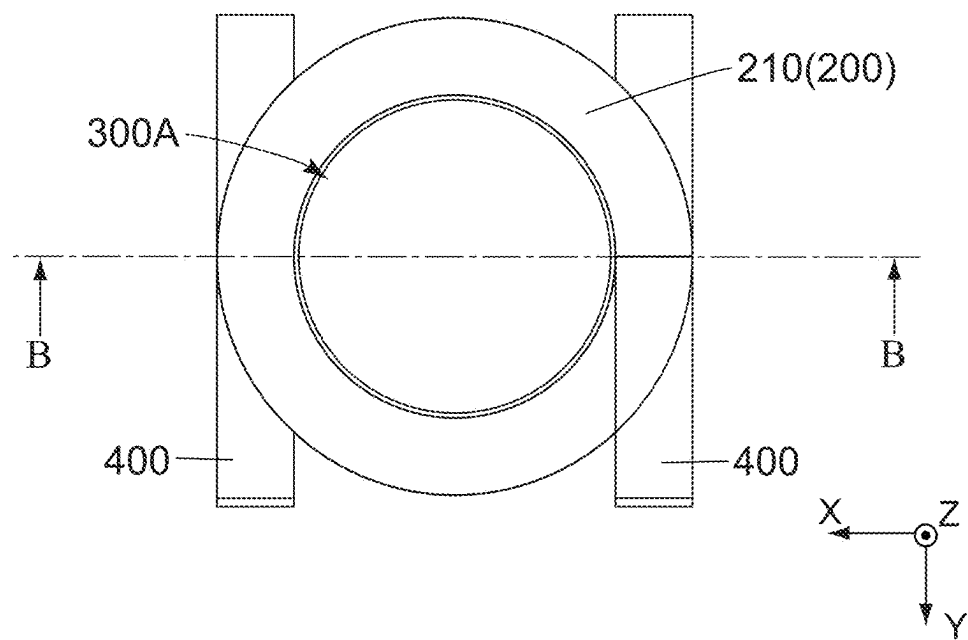


FIG. 19

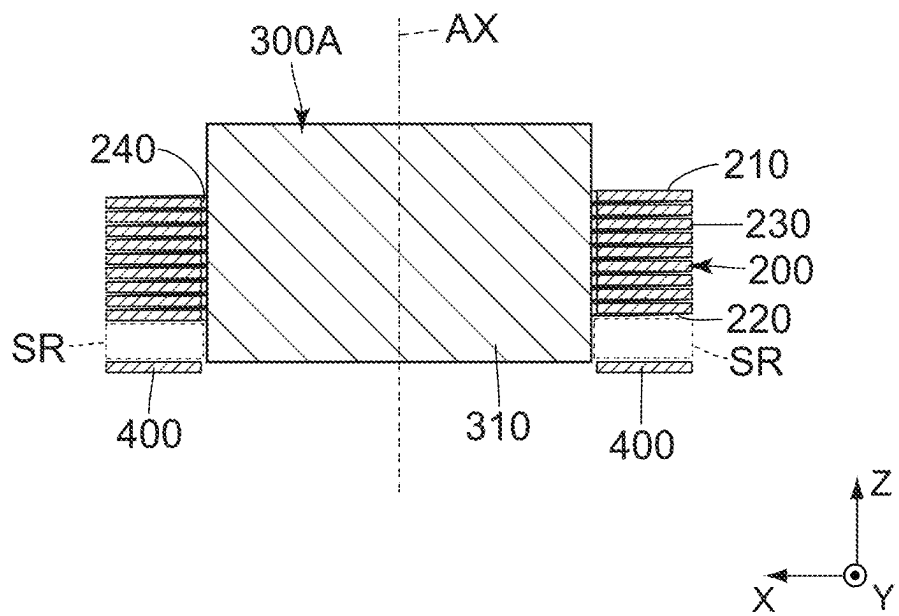


FIG. 20

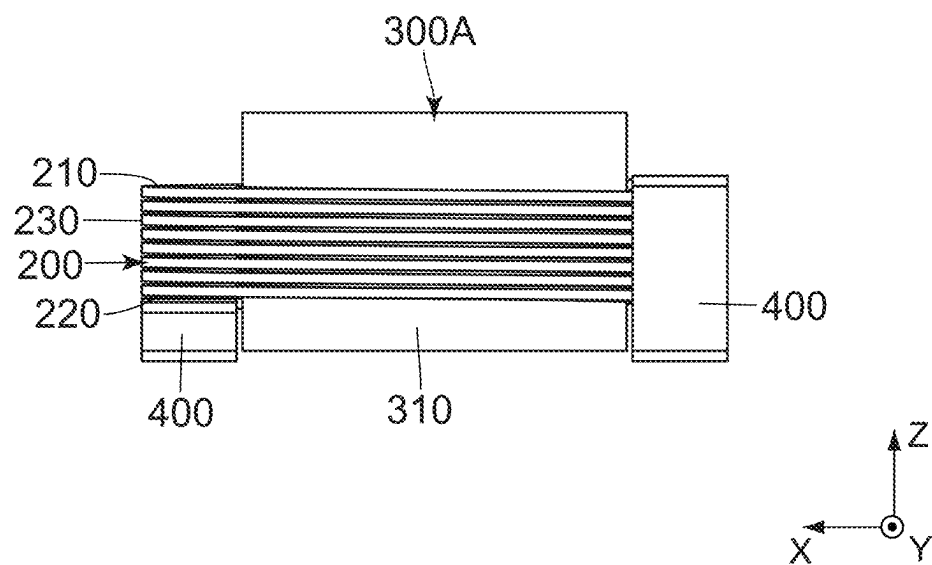


FIG. 21

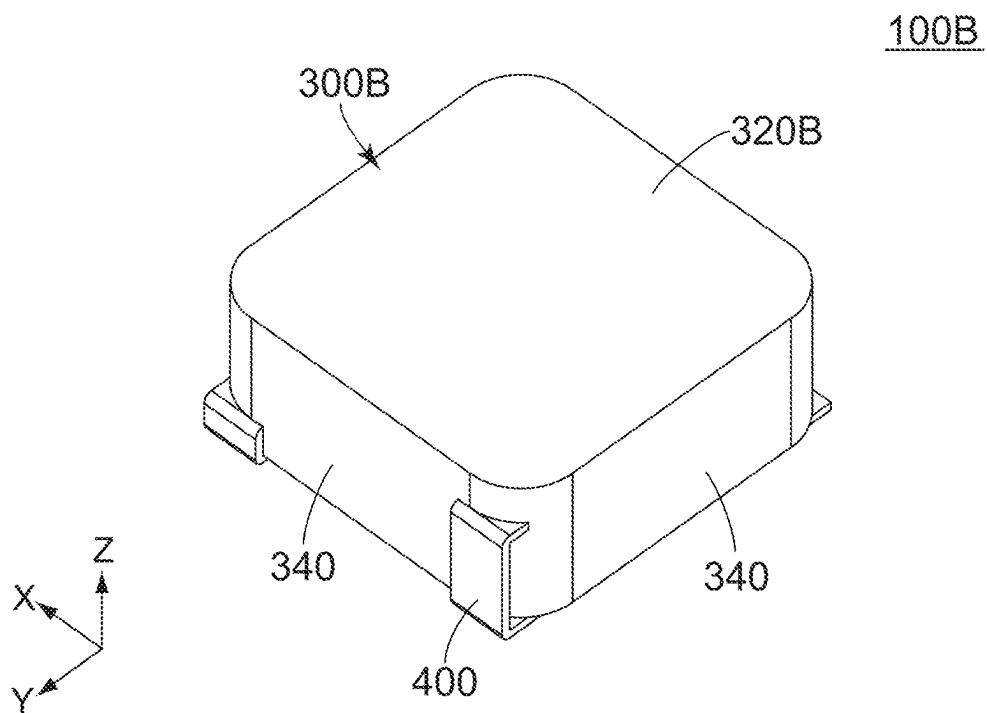


FIG. 22

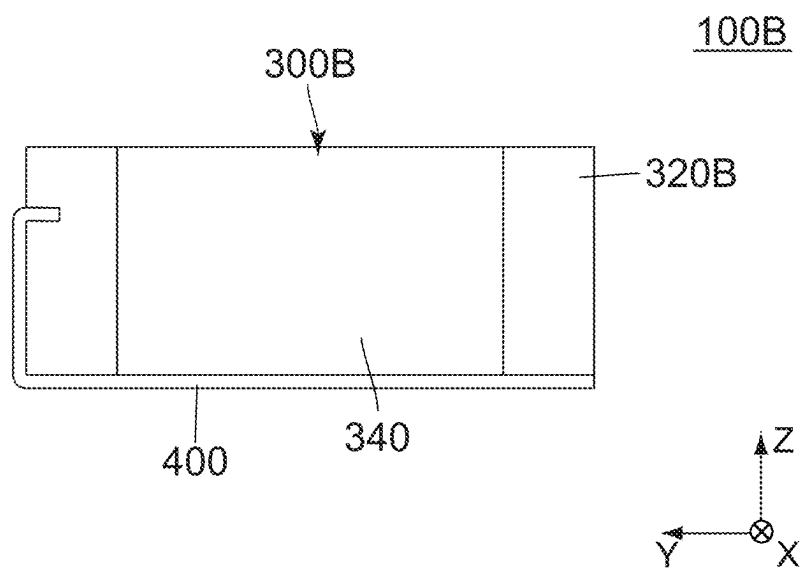


FIG. 23

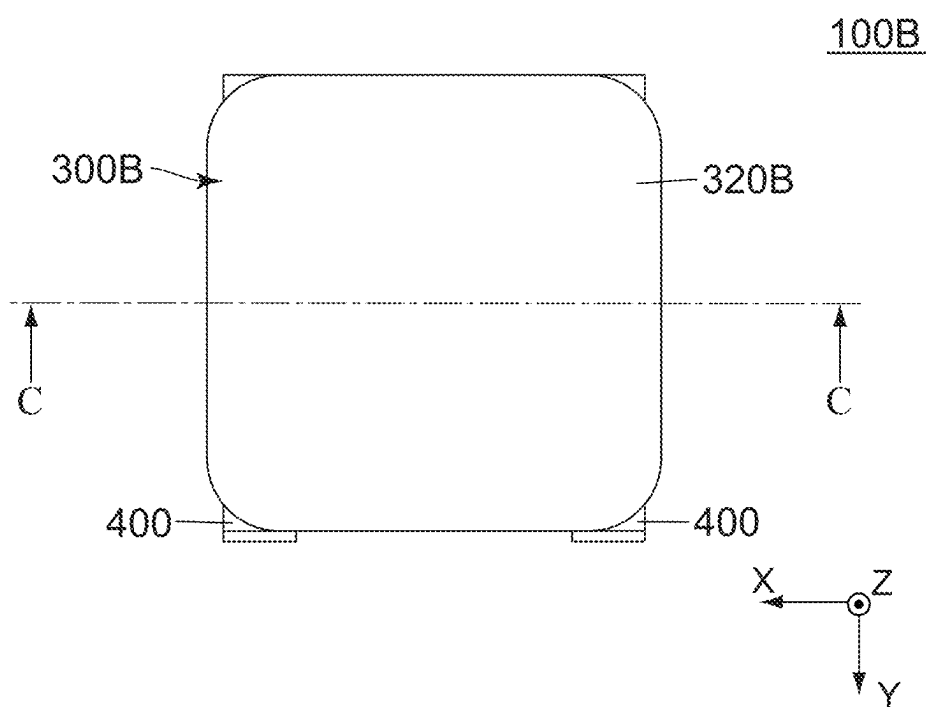


FIG. 24

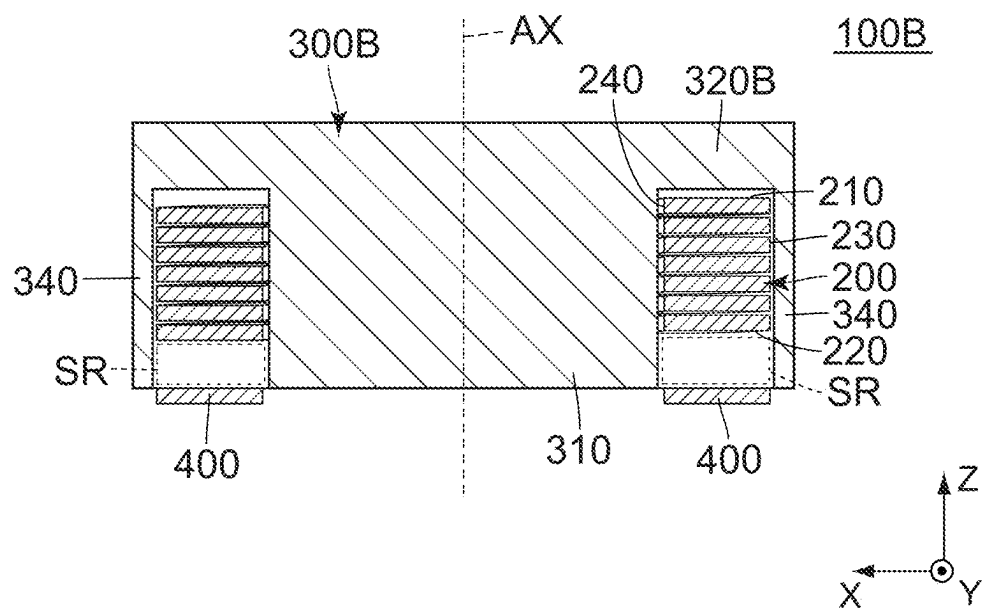


FIG. 25

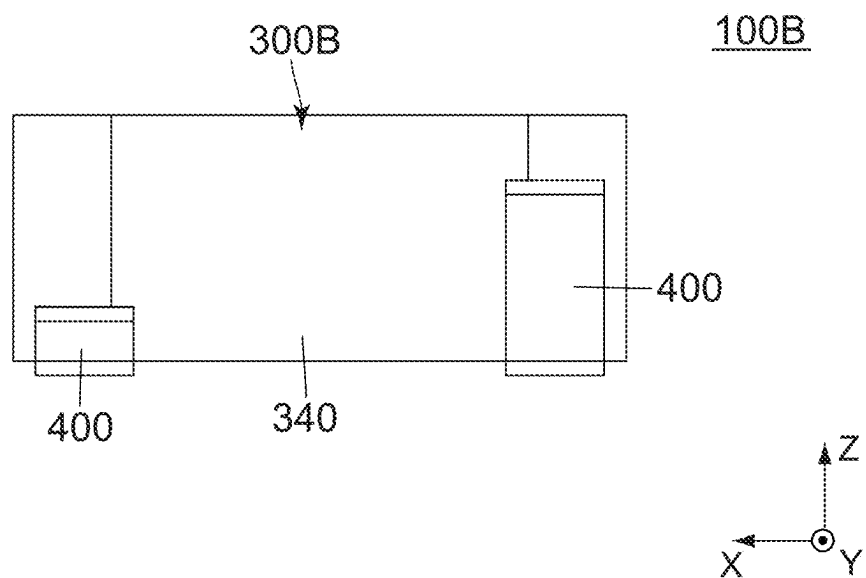


FIG. 26

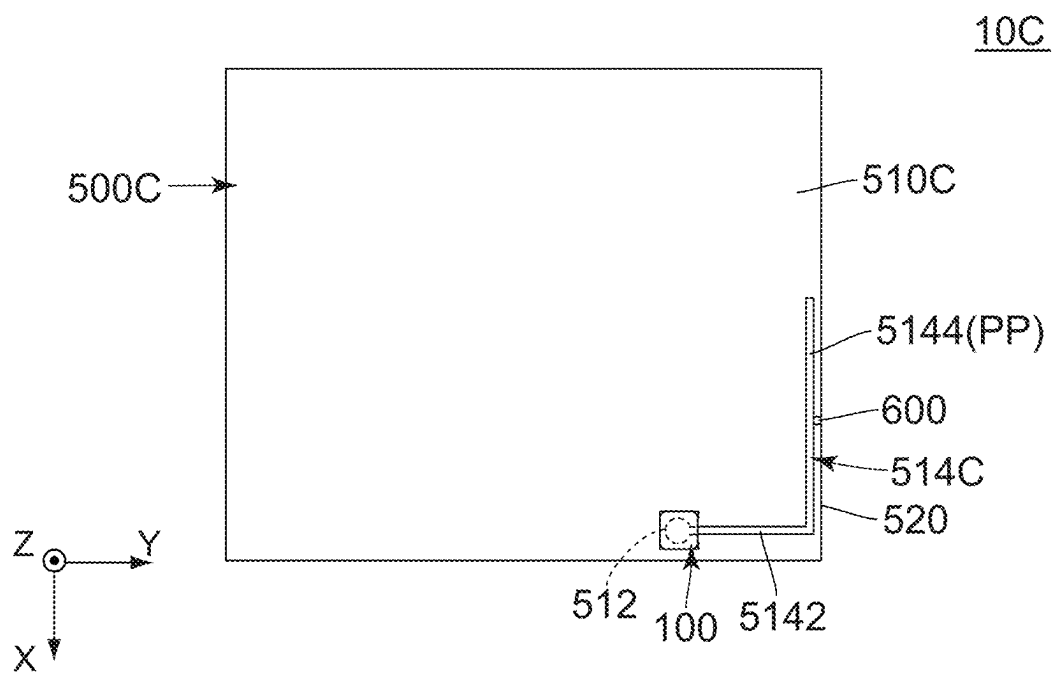


FIG. 27

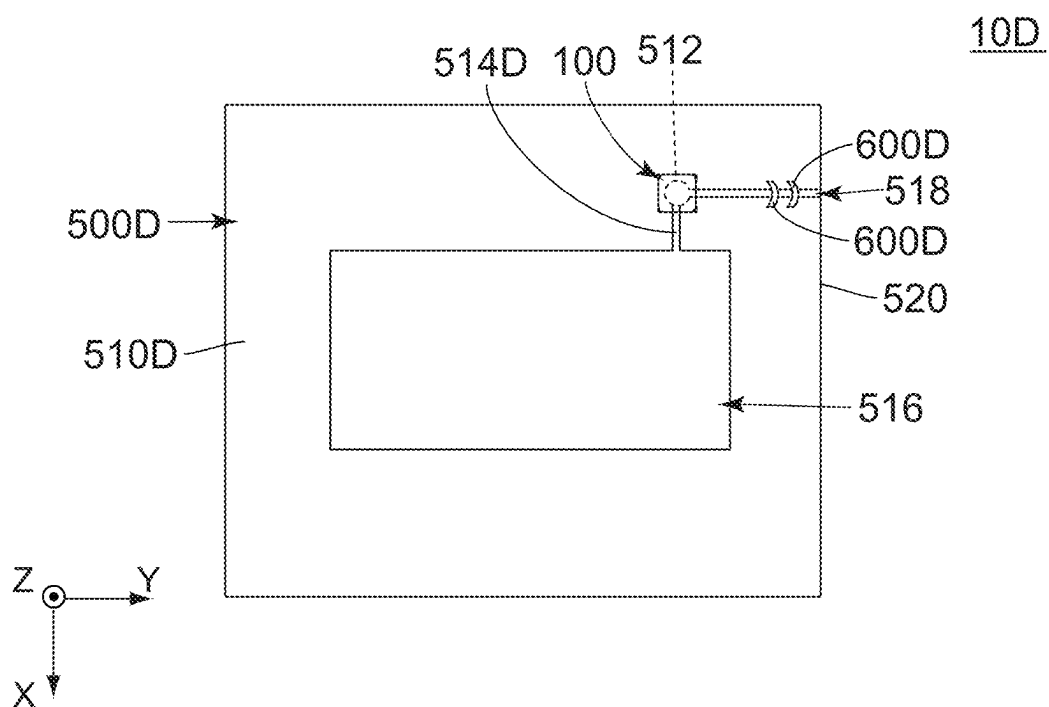
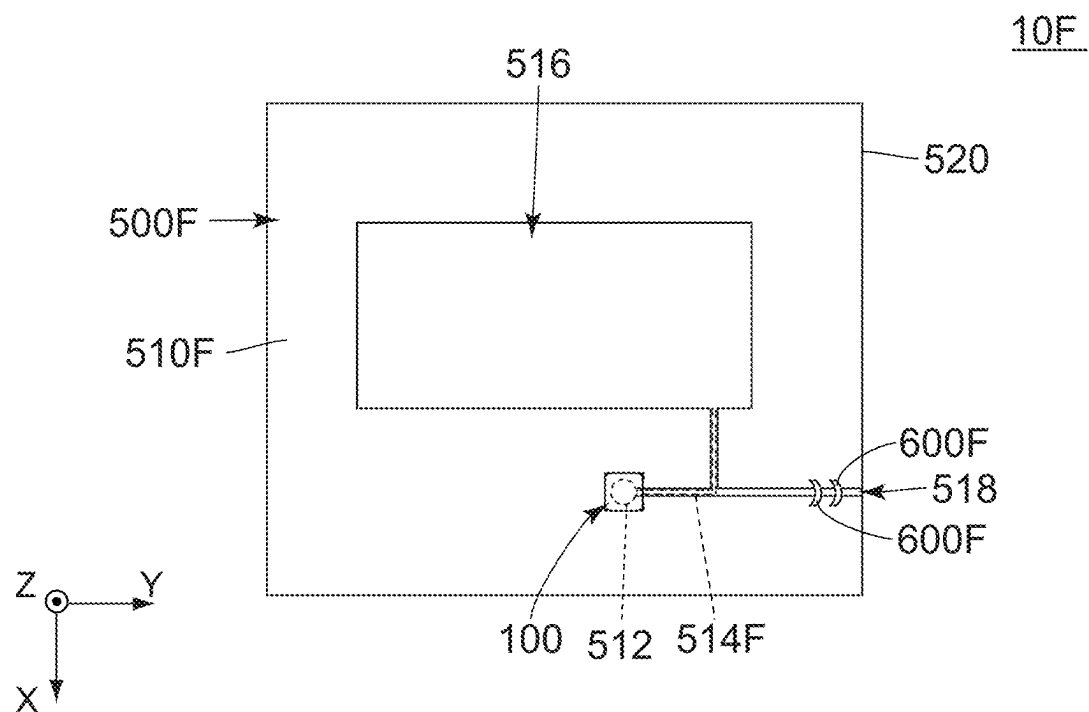
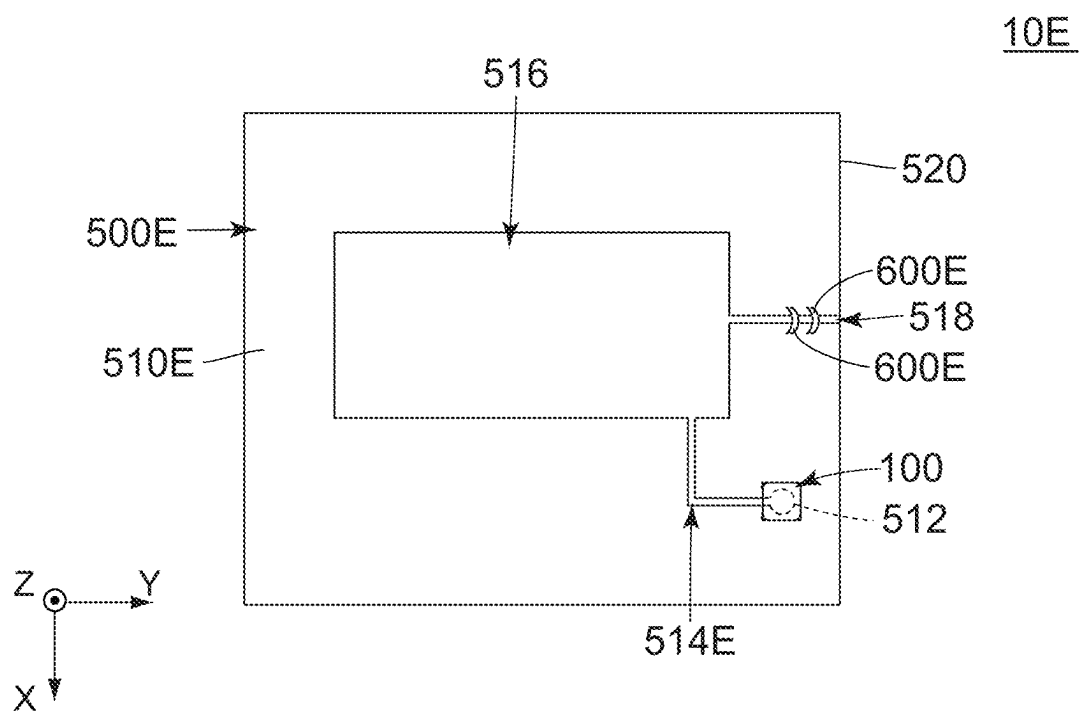
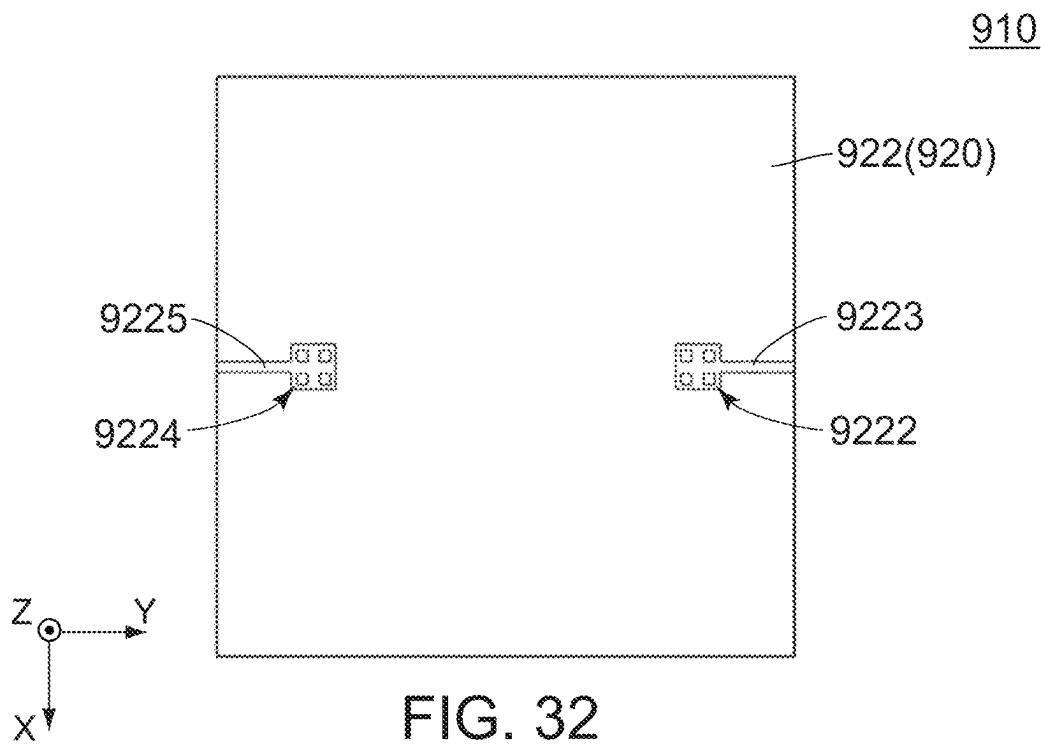
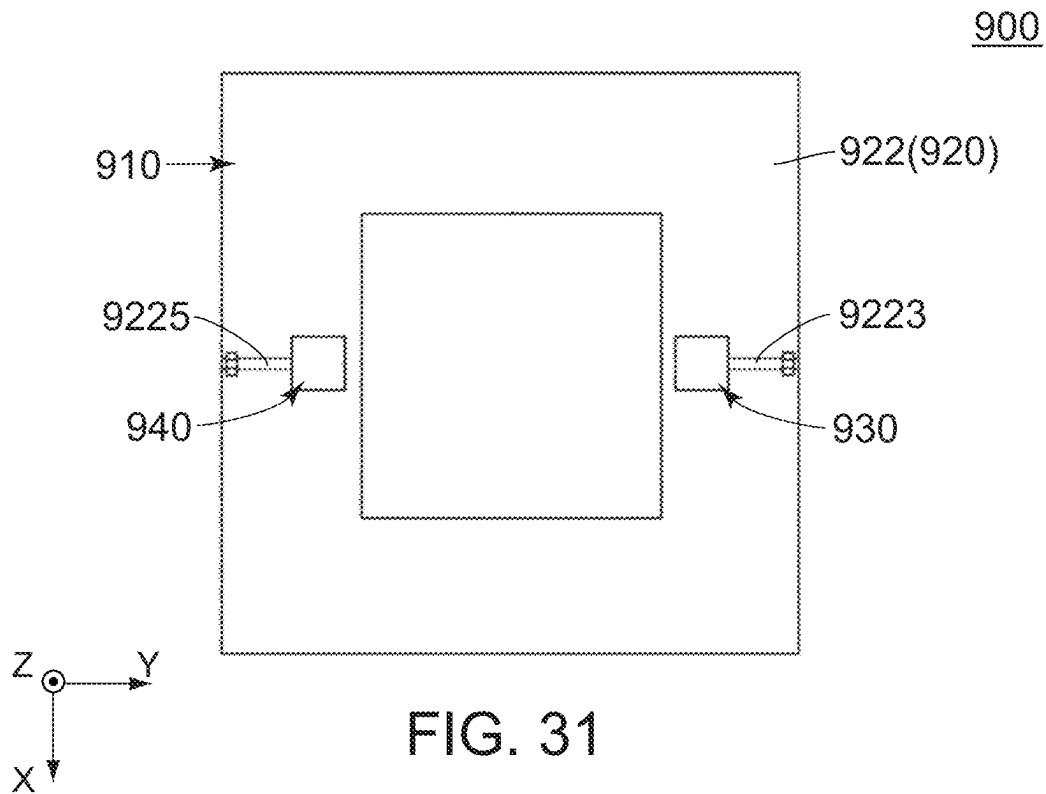


FIG. 28





ANTENNA DEVICE**TECHNICAL FIELD**

[0001] This invention relates to an antenna device comprising a circuit board and a coil antenna.

BACKGROUND ART

[0002] As shown in FIG. 31, an antenna device **900** of this type is disclosed in Patent Document 1. The antenna device **900** can be used for communications such as NFC (Near field communication) or RFID (radio frequency identifier). The antenna device **900** comprises a circuit board **910**, a first coil antenna (coil antenna) **930** and a second coil antenna (coil antenna) **940**. As shown in FIG. 32, a first principal surface **920** of the circuit board **910** is formed with a planar conductor (conductive portion) **922**. The conductive portion **922** is formed with a first opening (opening) **9222**, a second opening (opening) **9224**, a first slit (slit) **9223** and a second slit (slit) **9225**. The slit **9223** is coupled with the opening **9222**. The slit **9225** is coupled with the opening **9224**. Referring to FIGS. 31 and 32, the coil antenna **930** is mounted on the circuit board **910** so that the coil antenna **930** overlaps with the opening **9222** when the coil antenna **930** is viewed along a direction perpendicular to the circuit board **910**, or along a Z-direction. The coil antenna **940** is mounted on the circuit board **910** so that the coil antenna **940** overlaps with the opening **9224** when the coil antenna **940** is viewed along the direction perpendicular to the circuit board **910**, or along the Z-direction. Each of the coil antennas **930**, **940** comprises a coil (not shown) and a laminated body (not shown). Specifically, the coil has a winding axis extending in a direction perpendicular to the first principal surface **920**, or in the Z-direction, and the laminated body is formed by laminating insulation base layers each made of magnetic ferrite or the like.

PRIOR ART DOCUMENTS**Patent Document(s)**

[0003] Patent Document 1: JP-B 6566184

SUMMARY OF INVENTION**Technical Problem**

[0004] An antenna device such as the antenna device **900** of Patent document 1 is required to have further enhanced antenna characteristics.

[0005] It is therefore an object of the present invention to provide an antenna device having further enhanced antenna characteristics.

Solution to Problem

[0006] In a coil antenna whose coil is completely embedded in a magnetic core, a closed magnetic path is formed by the magnetic core. The inventors of the present invention have noticed that, when an antenna device such as the antenna device **900** is formed by such a coil antenna, magnetic flux generated by a coil passes in a closed magnetic path formed by a magnetic core and this weakens coupling of a current-loop, which flows at an outer periphery of a slit of a circuit board, with the coil antenna. Then, the inventors of the present invention have conceived that an

antenna device is configured to have an open magnetic path, through which magnetic flux generated by a coil of a coil antenna passes, by a magnetic core being at least not positioned just below the coil of the coil antenna. The inventors of the present invention have found that this new configuration can strengthen coupling of a current-loop, which flows at an outer periphery of a slit of a circuit board, with the coil antenna. The present invention is based on this finding.

[0007] An aspect of the present invention provides, as a first antenna device, an antenna device comprising a circuit board, a coil antenna and a resonance circuit, wherein:

[0008] the circuit board has a conductive portion with a flat shape;

[0009] the conductive portion is formed with an opening and a slit;

[0010] the conductive portion has a first path which goes around the opening and the slit;

[0011] the first path forms a part of the resonance circuit;

[0012] the slit is coupled with the opening;

[0013] the coil antenna is mounted on the circuit board so that the coil antenna at least overlaps with the opening when the coil antenna is viewed along an up-down direction perpendicular to the circuit board;

[0014] the coil antenna comprises a coil and a magnetic core;

[0015] the coil has a winding axis extending in the up-down direction;

[0016] the magnetic core partially forms a magnetic path of the coil;

[0017] the magnetic core is positioned at a region excluding a specific region which is positioned just below the coil;

[0018] the magnetic core at least has a center core; and

[0019] the center core is at least partially positioned inside the coil.

Advantageous Effects of Invention

[0020] The antenna device of the present invention is configured as follows: the coil antenna is mounted on the circuit board so that the coil antenna at least overlaps with the opening when the coil antenna is viewed along the up-down direction perpendicular to the circuit board; the magnetic core is positioned at the region excluding the specific region which is positioned just below the coil; the magnetic core at least has the center core; and the center core is at least partially positioned inside the coil. This strengthens coupling of a current-loop, which flows at an outer periphery of the slit of the circuit board, with the coil antenna in the antenna device of the present invention. In other words, the antenna device of the present invention has further enhanced antenna characteristics.

[0021] An appreciation of the objectives of the present invention and a more complete understanding of its structure may be had by studying the following description of the preferred embodiment and by referring to the accompanying drawings.

BRIEF DESCRIPTION OF DRAWINGS

[0022] FIG. 1 is a perspective view showing an antenna device according to a first embodiment of the present invention.

[0023] FIG. 2 is a top view showing the antenna device of FIG. 1.

[0024] FIG. 3 is a cross-sectional view showing a part of the antenna device of FIG. 2, taken along line A-A. In the figure, a molded resinous member, terminals and pads are omitted.

[0025] FIG. 4 is a perspective view showing a coil antenna which is included in the antenna device of FIG. 1.

[0026] FIG. 5 is a side view showing the coil antenna of FIG. 4.

[0027] FIG. 6 is a top view showing the coil antenna of FIG. 4.

[0028] FIG. 7 is a front view showing the coil antenna of FIG. 4.

[0029] FIG. 8 is a perspective view showing a state where the molded resinous member is removed from the coil antenna of FIG. 4.

[0030] FIG. 9 is a side view showing the coil antenna of FIG. 8.

[0031] FIG. 10 is a top view showing the coil antenna of FIG. 8.

[0032] FIG. 11 is a front view showing the coil antenna of FIG. 8.

[0033] FIG. 12 is a top view showing a circuit board which is included in the antenna device of FIG. 2.

[0034] FIG. 13 is a perspective view showing a first modification of the coil antenna of FIG. 4.

[0035] FIG. 14 is a side view showing the coil antenna of FIG. 13.

[0036] FIG. 15 is a top view showing the coil antenna of FIG. 13.

[0037] FIG. 16 is a front view showing the coil antenna of FIG. 13.

[0038] FIG. 17 is a perspective view showing a state where a molded resinous member is removed from the coil antenna of FIG. 13.

[0039] FIG. 18 is a side view showing the coil antenna of FIG. 17.

[0040] FIG. 19 is a top view showing the coil antenna of FIG. 17.

[0041] FIG. 20 is a cross-sectional view showing the coil antenna of FIG. 19, taken along line B-B.

[0042] FIG. 21 is a front view showing the coil antenna of FIG. 17.

[0043] FIG. 22 is a perspective view showing a second modification of the coil antenna of FIG. 4.

[0044] FIG. 23 is a side view showing the coil antenna of FIG. 22.

[0045] FIG. 24 is a top view showing the coil antenna of FIG. 22.

[0046] FIG. 25 is a cross-sectional view showing the coil antenna of FIG. 24, taken along line C-C. In the figure, a molded resinous member is omitted.

[0047] FIG. 26 is a front view showing the coil antenna of FIG. 22.

[0048] FIG. 27 is a top view showing an antenna device according to a second embodiment of the present invention.

[0049] FIG. 28 is a top view showing an antenna device according to a third embodiment of the present invention.

[0050] FIG. 29 is a top view showing an antenna device according to a fourth embodiment of the present invention.

[0051] FIG. 30 is a top view showing an antenna device according to a fifth embodiment of the present invention.

[0052] FIG. 31 is a top view showing an antenna device of Patent Document 1.

[0053] FIG. 32 is a top view showing a circuit board which is included in the antenna device of FIG. 31.

DESCRIPTION OF EMBODIMENTS

[0054] While the invention is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description thereto are not intended to limit the invention to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the present invention as defined by the appended claims.

First embodiment

[0055] 20 As shown in FIG. 1, an antenna device 10 according to a first embodiment of the present invention comprises a circuit board 500, a coil antenna 100 and a resonance circuit.

[0056] As shown in FIG. 12, the circuit board 500 of the present embodiment has a conductive portion 510 with a flat shape. In more detail, the conductive portion 510 is a ground conductor. Specifically, the circuit board 500 is configured so that a conductive layer, which functions as the conductive portion 510, is laminated on an insulating substrate such as a glass epoxy substrate.

[0057] As shown in FIG. 12, the conductive portion 510 is formed with an opening 512 and a slit 514. The conductive portion 510 has a first path which goes around the opening 512 and the slit 514. The first path forms a part of the resonance circuit.

[0058] Referring to FIG. 12, the opening 512 of the present embodiment is a part of the circuit board 500 which has no conductive pattern. The opening 512 has an outer edge which has a substantially circular shape when the circuit board 500 is viewed from above in an up-down direction. However, the present invention is not limited thereto, but the opening 512 may have an outer edge with a shape other than the substantially circular shape. In the present embodiment, the up-down direction is a Z-direction. Specifically, it is assumed that upward is a positive Z-direction while downward is a negative Z-direction.

[0059] As shown in FIG. 12, similar to the opening 512, the slit 514 of the present embodiment is a part of the circuit board 500 which has no conductive pattern. The slit 514 is coupled with the opening 512. The slit 514 extends from the opening 512. In the present embodiment, the number of the slit 514 extending from the opening 512 is one. However, the present invention is not limited thereto, but a plurality of the slits 514 may extend from the opening 512. The slit 514 does not communicate with the outside of the conductive portion 510 in a plane perpendicular to the up-down direction. The slit 514 has a predetermined portion PP. The slit 514 has a first portion 5142 and a second portion 5144.

[0060] As shown in FIG. 12, a width of the slit 514 is smaller than a diameter of the opening 512. However, the present invention is not limited thereto, but the width of the slit 514 may be equal to the diameter of the opening 512, or the width of the slit 514 may be greater than the diameter of the opening 512. It is noted that an increase of the width of

the slit 514 enlarges a current loop, which is formed around the slit 514, so that a communication range of the antenna device 10 is increased.

[0061] As shown in FIG. 12, the first portion 5142 extends rightward in a right-left direction from the opening 512. In the present embodiment, the right-left direction is a Y-direction. Specifically, rightward is a positive Y-direction while leftward is a negative Y-direction.

[0062] As shown in FIG. 12, the second portion 5144 extends from the first portion 5142 in a direction intersecting with the first portion 5142. The second portion 5144 extends rearward in a front-rear direction from a right end of the first portion 5142. In the present embodiment, the front-rear direction is an X-direction. Specifically, forward is a positive X-direction while rearward is a negative X-direction. The second portion 5144 functions as the predetermined portion PP.

[0063] As shown in FIG. 12, the slit 514 further comprises a third portion 5146.

[0064] As shown in FIG. 12, the third portion 5146 extends from the second portion 5144 in a direction intersecting with the second portion 5144. The third portion 5146 extends leftward in the right-left direction from a rear end of the second portion 5144.

[0065] As shown in FIG. 12, the circuit board 500 has an edge 520.

[0066] As shown in FIG. 12, the edge 520 defines an outer edge of the circuit board 500. Specifically, the edge 520 defines the outer edge of the circuit board 500 in a direction perpendicular to the up-down direction. The slit 514 extends from the opening 512 toward the edge 520, and is bent so that it extends along the edge 520, and is further bent so that it extends away from the edge 520. The slit 514 does not reach the edge 520. The first portion 5142 extends from the opening 512 toward the edge 520. The predetermined portion PP extends along the edge 520 in the vicinity of the edge 520. This configuration enables the circuit board 500 to have an increased region where electronic components can be arranged on the circuit board 500, and the communication range of the antenna device 10 can be increased by enlarging the current loop, which is formed around the slit 514, because this configuration can increase a length of the slit 514.

[0067] As described above, the slit 514 extends from the opening 512 toward the edge 520, and is bent so that it extends along the edge 520, and is further bent so that it extends away from the edge 520. Additionally, as described above, the slit 514 does not reach the edge 520 and does not communicate with the outside of the conductive portion 510 in the plane perpendicular to the up-down direction. However, the present invention is not limited thereto, but the slit 514 may extend toward the edge 520 and may reach the edge 520. Specifically, the slit 514 may communicate with the outside of the conductive portion 510 in the plane perpendicular to the up-down direction.

[0068] Referring to FIGS. 2 and 12, the coil antenna 100 is mounted on the circuit board 500 so that the coil antenna 100 at least overlaps with the opening 512 when the coil antenna 100 is viewed along the up-down direction perpendicular to the circuit board 500. The coil antenna 100 is surface-mounted on the circuit board 500. In other words, the coil antenna 100 is a surface mount antenna.

[0069] As shown in FIG. 3, the coil antenna 100 comprises a coil 200 and a magnetic core 300.

[0070] As shown in FIG. 3, the coil 200 of the present embodiment has a winding axis AX which extends in the up-down direction. This coil 200 is formed by winding a jacketed flat electrical cable edgewise. However, the present invention is not limited thereto, but the coil 200 may be formed by winding a jacketed round electrical cable, and a material of the conductive cable, a cross-section of the conductive cable and a manner of winding the conductive cable are not limited to the present embodiment. The coil 200 has a ring shape when viewed along the up-down direction. However, the present invention is not limited thereto, but the coil 200 may have a shape other than the ring shape. The coil 200 has an upper end 210 and a lower end 220 in the up-down direction. The coil 200 has an inner periphery 240 and an outer periphery 230.

[0071] Referring to FIGS. 3 and 12, S1 and S2 satisfy $1 \leq S2/S1 \leq 2$, where S1 is a cross-sectional area of the inner periphery 240 of the coil 200, and S2 is an area of the opening 512. This can maximize a coupling coefficient between the coil antenna 100 and the slit 514, and thereby the coil antenna 100 can easily be electromagnetically coupled with a current (current loop) which goes around the opening 512 and the slit 514 of the conductive portion 510.

[0072] Referring to FIG. 3, the magnetic core 300 of the present embodiment is a cast core. More specifically, the magnetic core 300 is a composite magnet in which soft magnetic particles made of ferrous alloy, ferrite or the like are dispersed in a hardened binder. In other words, the magnetic core 300 is formed by hardening a slurry which includes the soft magnetic particles and the binder or the like. It is noted that a method of manufacturing the magnetic core 300 is not limited to the aforementioned method and that the magnetic core 300 may be formed by any manufacturing method. The magnetic core 300 partially forms a magnetic path of the coil 200. The magnetic core 300 is positioned at a region excluding a specific region SR which is positioned just below the coil 200. In other words, the magnetic core 300 has no portion which is positioned just below the coil 200. It is noted that the specific region SR is a region which is hidden below the coil 200 when the coil 200 is viewed alone from above. The specific region SR is positioned below the lower end 220 of the coil 200 in the up-down direction. The specific region SR is positioned outside the inner periphery 240 of the coil 200 in a radial direction perpendicular to the winding axis AX. The specific region SR is positioned inside the outer periphery 230 of the coil 200 in the radial direction perpendicular to the winding axis AX. The magnetic core 300 at least has a center core 310.

[0073] Referring to FIG. 3, the center core 310 of the present embodiment has a cylindrical shape extending in the up-down direction. The center core 310 is at least partially positioned inside the coil 200. A center axis of the cylindrical shape of the center core 310 is coincide with the winding axis AX of the coil 200. A size of a cross-section, which is perpendicular to the up-down direction, of the center core 310 is smaller than the diameter of the opening 512. A projected image of the center core 310, which is projected onto the circuit board 500 in the up-down direction, is positioned within the opening 512.

[0074] As described above, the coil antenna 100 is mounted on the circuit board 500 so that the coil antenna 100 at least overlaps with the opening 512 when the coil antenna 100 is viewed along the up-down direction perpendicular to

the circuit board 500, the magnetic core 300 is positioned at the region excluding the specific region SR which is positioned just below the coil 200, the magnetic core 300 at least has the center core 310 and the center core 310 is at least partially positioned inside the coil 200. Thus, in the antenna device 10 of the present embodiment, a part of a flux of the coil 200 flows to the opening 512 of the circuit board 500 and thereby the coil antenna 100 can easily be electromagnetically coupled with the current (current loop) which goes around the opening 512 and the slit 514 of the conductive portion 510. In other words, this strengthens coupling of the current-loop, which flows at an outer periphery of the slit 514 of the circuit board 500, with the coil antenna 100 in the antenna device 10 of the present embodiment, and thereby the antenna device 10 of the present embodiment has further enhanced antenna characteristics.

[0075] As shown in FIG. 3, the center core 310 extends downward in the up-down direction beyond the lower end 220 of the coil 200. Thus, in the antenna device 10 of the present embodiment, the part of the flux of the coil 200 further flows to the opening 512 of the circuit board 500 and thereby the coil antenna 100 can more easily be electromagnetically coupled with the current (current loop) which goes around the opening 512 and the slit 514 of the conductive portion 510. In other words, this further strengthens the coupling of the current-loop, which flows at the outer periphery of the slit 514 (see FIG. 12) of the circuit board 500, with the coil antenna 100 in the antenna device 10 of the present embodiment, and thereby the antenna device 10 of the present embodiment has more further enhanced antenna characteristics.

[0076] As shown in FIG. 3, the center core 310 extends upward in the up-down direction beyond the upper end 210 of the coil 200.

[0077] As shown in FIG. 3, the magnetic core 300 further has an upper portion 320.

[0078] As shown in FIG. 3, the upper portion 320 extends from the center core 310 toward the outer periphery 230 of the coil 200. In the radial direction perpendicular to the center axis of the cylindrical shape of the center core 310, an outer end of the upper portion 320 is positioned outside the outer periphery 230 of the coil 200. The upper portion 320 overlaps with the coil 200 when the upper portion 320 is viewed in the up-down direction. This enables the flux of the coil 200, which flows through the center core 310 to an upper part of the magnetic core 300, to efficiently flow toward the outer periphery 230 of the coil 200.

[0079] As shown in FIG. 7, the coil antenna 100 further has two terminals 400.

[0080] Referring to FIG. 8, the terminals 400 of the present embodiment are attached to end portions, respectively, of the coil 200. However, the present invention is not limited thereto, but the terminal 400 may be a portion formed by elongating the end portion of the coil 200. Each of the terminals 400 defines a lower end of the coil antenna 100. The terminals 400 are connected to a communication circuit (not shown) via pads of the circuit board 500.

[0081] As shown in FIG. 4, the coil antenna 100 comprises a molded resinous member 700.

[0082] Referring to FIGS. 5 and 9, the molded resinous member 700 covers the coil 200. Referring to FIGS. 3, 5 and 7, the specific region SR is occupied by the molded resinous member 700. In other words, the molded resinous member 700 is positioned just below the coil 200.

[0083] As shown in FIG. 2, the antenna device 10 further has a capacitor 600 which forms a part of the resonant circuit.

[0084] Referring to FIG. 12, the capacitor 600 forms a current loop path together with the first path which goes around the opening 512 and the slit 514 of the conductive portion 510. In other words, the capacitor 600 forms the resonant circuit together with the first path which goes around the opening 512 and the slit 514 of the conductive portion 510. A resonant frequency of the resonant circuit is a communication frequency band of the antenna device 10. The capacitor 600 is mounted on the circuit board 500 at its region between the edge 520 and the slit 514. In more detail, the capacitor 600 is mounted on the circuit board 500 at its region between the edge 520 and the second portion 514A. However, the mounting location of the capacitor 600 is not limited to that described above, but the capacitor 600 may be mounted at any location on the conductive portion 510, provided that the capacitor 600 and the first path, which goes around the opening 512 and the slit 514 of the conductive portion 510, form the current loop path.

[0085] The configuration of the coil antenna 100 is not limited thereto, but the coil antenna 100 can be modified, for example, as described below.

First Modification

[0086] Referring to FIGS. 13 and 20, a coil antenna 100A according to a first modification of the present invention comprises a coil 200 and a magnetic core 300A. The coil 200 of the present modification has a structure same as that of the coil 200 of the aforementioned embodiment, and detail explanation thereabout will be omitted.

[0087] Referring to FIG. 20, the magnetic core 300A of the present embodiment is a cast core similar to the aforementioned magnetic core 300. The magnetic core 300A partially forms a magnetic path of the coil 200. The magnetic core 300A is positioned at a region excluding a specific region SR which is positioned just below the coil 200. It is noted that the specific region SR is a region which is hidden below the coil 200 when the coil 200 is viewed from above. The specific region SR is positioned below a lower end 220 of the coil 200 in the up-down direction. The specific region SR is positioned outside an inner periphery 240 of the coil 200 in a radial direction perpendicular to a winding axis AX. The specific region SR is positioned inside an outer periphery 230 of the coil 200 in the radial direction perpendicular to the winding axis AX. The magnetic core 300A has a center core 310.

[0088] Referring to FIGS. 17 and 20, the center core 310 of the present embodiment has a cylindrical shape extending in the up-down direction. The center core 310 is at least partially positioned inside the coil 200.

[0089] As shown in FIG. 20, the center core 310 extends downward in the up-down direction beyond a lower end 220 of the coil 200. The center core 310 extends upward in the up-down direction beyond an upper end 210 of the coil 200.

[0090] Referring to FIGS. 10 and 20, dissimilar to magnetic core 300, the magnetic core 300A has no upper portion 320.

[0091] As shown in FIG. 13, the coil antenna 100A comprises a molded resinous member 700A.

[0092] Referring to FIGS. 14 and 18, the molded resinous member 700A covers the coil 200. Referring to FIG. 20, the specific region SR is occupied by the molded resinous

member **700A**. In other words, the molded resinous member **700A** is positioned just below the coil **200**. It is noted that an upper end of the center core **310** is not covered with the molded resinous member **700A** as referred to FIG. **15**.

[0093] An antenna device (not shown), which is configured so that the coil antenna **100A** of the present modification is mounted on a circuit board **500**, can also have the same effects as the aforementioned embodiment.

Second Modification

[0094] Referring to FIGS. **22** and **25**, a coil antenna **100B** according to a second modification of the present invention comprises a coil **200** and a magnetic core **300B**. The coil **200** of the present modification has a structure same as that of the coil **200** of the aforementioned embodiment, and detail explanation thereabout will be omitted.

[0095] Referring to FIG. **25**, the magnetic core **300B** of the present embodiment is a cast core similar to the aforementioned magnetic core **300**. The magnetic core **300B** partially forms a magnetic path of the coil **200**. The magnetic core **300B** is positioned at a region excluding a specific region **SR** which is positioned just below the coil **200**. It is noted that the specific region **SR** is a region which is hidden below the coil **200** when the coil **200** is viewed from above. The specific region **SR** is positioned below a lower end **220** of the coil **200** in the up-down direction. The specific region **SR** is positioned outside an inner periphery **240** of the coil **200** in a radial direction perpendicular to a winding axis **AX**. The specific region **SR** is positioned inside an outer periphery **230** of the coil **200** in the radial direction perpendicular to the winding axis **AX**. The magnetic core **300B** at least has a center core **310**.

[0096] Referring to FIG. **25**, the center core **310** of the present embodiment has a cylindrical shape extending in the up-down direction. The center core **310** is at least partially positioned inside the coil **200**.

[0097] As shown in FIG. **25**, the center core **310** extends downward in the up-down direction beyond a lower end **220** of the coil **200**. The center core **310** extends upward in the up-down direction beyond an upper end **210** of the coil **200**.

[0098] As shown in FIG. **25**, the magnetic core **300B** further has an upper portion **320B**. The upper portion **320B** of the present modification has a structure same as that of the upper portion **320** of the aforementioned embodiment, and detail explanation thereabout will be omitted.

[0099] As shown in FIG. **25**, the magnetic core **300B** further has a side portion **340**.

[0100] As shown in FIG. **25**, the side portion **340** of the present embodiment is positioned outside the coil **200** in a plane perpendicular to the up-down direction. The side portion **340** extends downward in the up-down direction from the upper portion **320B**. This enables a flux of the coil **200**, which passes through the upper portion **320B**, to efficiently flow toward a lower part of the coil **200**. The side portion **340** covers the whole of the outer periphery **230** of the coil **200**. However, the present invention is not limited thereto, but the side portion **340** may cover only a part of the outer periphery **230** of the coil **200**.

[0101] Referring to FIG. **25**, the coil antenna **100B** comprises a molded resinous member (not shown).

[0102] Referring to FIG. **25**, the molded resinous member of the present modification covers the coil **200**. The specific

region **SR** is occupied by the molded resinous member. In other words, the molded resinous member is positioned just below the coil **200**.

[0103] An antenna device (not shown), which is configured so that the coil antenna **100B** of the present modification is mounted on a circuit board **500**, can also have the same effects as the aforementioned embodiment.

Second Embodiment

[0104] As shown in FIG. **27**, an antenna device **10C** according to a second embodiment of the present invention comprises a circuit board **500C**, a coil antenna **100** and a resonance circuit. The coil antenna **100** of the present embodiment has a structure same as that of the coil antenna **100** of the first embodiment, and detail explanation thereabout will be omitted. As for directions and orientations in the present embodiment, expressions same as those of the first embodiment will be used hereinbelow.

[0105] As shown in FIG. **27**, the circuit board **500C** of the present embodiment has a conductive portion **510C** with a flat shape. In more detail, the conductive portion **510C** is a ground conductor. Specifically, the circuit board **500C** is configured so that a conductive layer, which functions as the conductive portion **510C**, is laminated on an insulating substrate such as a glass epoxy substrate.

[0106] As shown in FIG. **27**, the conductive portion **510C** is formed with an opening **512** and a slit **514C**. The opening **512** of the present embodiment has a structure same as that of the opening **512** of the first embodiment, and detail explanation thereabout will be omitted. The conductive portion **510C** has a first path which goes around the opening **512** and the slit **514C**. The first path forms a part of the resonance circuit.

[0107] As shown in FIG. **27**, similar to the opening **512**, the slit **514C** of the present embodiment is a part of the circuit board **500C** which has no conductive pattern. The slit **514C** is coupled with the opening **512**. The slit **514C** extends from the opening **512**. The slit **514C** does not communicate with the outside of the conductive portion **510C** in a plane perpendicular to the up-down direction. The slit **514C** has a predetermined portion **PP**. The slit **514C** has a first portion **5142** and a second portion **5144**.

[0108] As shown in FIG. **27**, a width of the slit **514C** is smaller than a diameter of the opening **512**. However, the present invention is not limited thereto, but the width of the slit **514C** may be equal to the diameter of the opening **512**, or the width of the slit **514C** may be greater than the diameter of the opening **512**. It is noted that an increase of the width of the slit **514C** enlarges a current loop, which is formed around the slit **514C**, so that a communication range of the antenna device **10C** is increased.

[0109] As shown in FIG. **27**, the first portion **5142** extends rightward in the right-left direction from the opening **512**.

[0110] As shown in FIG. **27**, the second portion **5144** extends from the first portion **5142** in a direction intersecting with the first portion **5142**. The second portion **5144** extends rearward in the front-rear direction from a right end of the first portion **5142**. The second portion **5144** functions as the predetermined portion **PP**.

[0111] Referring to FIGS. **2** and **27**, dissimilar to the slit **514** of the first embodiment, the slit **514C** does not comprise a third portion **5146**.

[0112] As shown in FIG. **27**, the circuit board **500C** has an edge **520**.

[0113] As shown in FIG. 27, the edge 520 of the present embodiment defines an outer edge of the circuit board 500C. Specifically, the edge 520 defines the outer edge of the circuit board 500C in a direction perpendicular to the up-down direction. The first portion 5142 extends from the opening 512 toward the edge 520. The predetermined portion PP extends along the edge 520 in the vicinity of the edge 520. This configuration enables the circuit board 500C to have an increased region where electronic components can be arranged on the circuit board 500C, and the communication range of the antenna device 10C can be increased by enlarging the current loop, which is formed around the slit 514C, because this configuration can increase a length of the slit 514C.

[0114] As shown in FIG. 27, the antenna device 10C further has a capacitor 600 which forms a part of the resonant circuit.

[0115] Referring to FIG. 27, the capacitor 600 forms a current loop path together

[0116] with the first path which goes around the opening 512 and the slit 514C of the conductive portion 510C. In other words, the capacitor 600 forms the resonant circuit together with the first path which goes around the opening 512 and the slit 514C of the conductive portion 510C. A resonant frequency of the resonant circuit is a communication frequency band of the antenna device 10C. As shown in FIG. 27, the capacitor 600 is mounted on the circuit board 500C at its region between the edge 520 and the slit 514C. In more detail, the capacitor 600 is mounted on the circuit board 500C at its region between the edge 520 and the second portion 5144. However, the mounting location of the capacitor 600 is not limited to that described above, but the capacitor 600 may be mounted at any location on the conductive portion 510C, provided that the capacitor 600 and the first path, which goes around the opening 512 and the slit 514C of the conductive portion 510C, form the current loop path.

Third Embodiment

[0117] As shown in FIG. 28, an antenna device 10D according to a third embodiment of the present invention comprises a circuit board 500D, a coil antenna 100 and a resonance circuit. The coil antenna 100 of the present embodiment has a structure same as that of the coil antenna 100 of the first embodiment, and detail explanation thereabout will be omitted. As for directions and orientations in the present embodiment, expressions same as those of the first embodiment will be used hereinbelow.

[0118] As shown in FIG. 28, the circuit board 500D of the present embodiment has a conductive portion 510D with a flat shape. In more detail, the conductive portion 510D is a ground conductor. Specifically, the circuit board 500D is configured so that a conductive layer, which functions as the conductive portion 510D, is laminated on an insulating substrate such as a glass epoxy substrate.

[0119] As shown in FIG. 28, the conductive portion 510D is formed with an opening 512 and a slit 514D. The conductive portion 510D has a first path which goes around the opening 512 and the slit 514D. The first path forms a part of the resonance circuit. The opening 512 of the present embodiment has a structure same as that of the opening 512 of the first embodiment, and detail explanation thereabout will be omitted.

[0120] As shown in FIG. 28, similar to the opening 512, the slit 514D of the present embodiment is a part of the circuit board 500D which has no conductive pattern. The slit 514D is coupled with the opening 512. The slit 514D extends from the opening 512. The slit 514D extends from the opening 512 in the front-rear direction. The slit 514D does not communicate with the outside of the conductive portion 510D in a plane perpendicular to the up-down direction.

[0121] As shown in FIG. 28, a width of the slit 514D is smaller than a diameter of the opening 512. However, the present invention is not limited thereto, but the width of the slit 514D may be equal to the diameter of the opening 512, or the width of the slit 514D may be greater than the diameter of the opening 512. It is noted that an increase of the width of the slit 514D enlarges a current loop, which is formed around the slit 514D, so that a communication range of the antenna device 10D is increased.

[0122] As shown in FIG. 28, the conductive portion 510D is further formed with an additional opening 516.

[0123] As shown in FIG. 28, the additional opening 516 of the present embodiment is a hole piercing the circuit board 500D. In other words, the additional opening 516 pierces the circuit board 500D in the up-down direction. However, the present invention is not limited thereto, but the additional opening 516 may be a part of the circuit board 500D, which has no conductive pattern, similar to the opening 512. The additional opening 516 is positioned forward of the opening 512 in the front-rear direction. In the plane perpendicular to the up-down direction, the additional opening 516 is greater than the opening 512. The conductive portion 510D has a second path which goes around the additional opening 516. The second path forms a part of the resonance circuit. The slit 514D is coupled with both of the opening 512 and the additional opening 516.

[0124] As shown in FIG. 28, the conductive portion 510D is further formed with an additional slit 518.

[0125] As shown in FIG. 28, the additional slit 518 of the present embodiment is a part of the circuit board 500D, which has no conductive pattern, similar to the opening 512. The additional slit 518 is coupled with the opening 512. The additional slit 518 extends from the opening 512. The additional slit 518 extends in the right-left direction. The additional slit 518 communicates with the outside of the conductive portion 510D in the plane perpendicular to the up-down direction. All of the opening 512, the slit 514D, the additional opening 516 and the additional slit 518 are coupled to one another.

[0126] As shown in FIG. 28, a width of the additional slit 518 is smaller than the diameter of the opening 512. However, the present invention is not limited thereto, but the width of the additional slit 518 may be equal to the diameter of the opening 512, or the width of the additional slit 518 may be greater than the diameter of the opening 512.

[0127] As shown in FIG. 28, the circuit board 500D has an edge 520.

[0128] As shown in FIG. 28, the edge 520 defines an outer edge of the circuit board 500. Specifically, the edge 520 defines the outer edge of the circuit board 500D in a direction perpendicular to the up-down direction. The slit 514D does not reach the edge 520. The additional slit 518 extends toward the edge 520. Specifically, the additional slit 518 extends from the opening 512 toward the edge 520. The additional slit 518 reaches the edge 520.

[0129] Referring to FIG. 28, the coil antenna 100 is mounted on the circuit board 500D so that the coil antenna 100 at least overlaps with the opening 512 when the coil antenna 100 is viewed along the up-down direction perpendicular to the circuit board 500D. The coil antenna 100 is mounted on the circuit board 500D so that the coil antenna 100 does not overlap with the additional opening 516 when the coil antenna 100 is viewed along the up-down direction. The coil antenna 100 is surface-mounted on the circuit board 500D.

[0130] As shown in FIG. 28, the antenna device 10D further has capacitors 600D which form a part of the resonant circuit.

[0131] Referring to FIG. 28, the capacitors 600D form a current loop path together with a path which goes around the opening 512, the additional opening 516, the slit 514D and the additional slit 518 of the conductive portion 510D. In other words, the capacitors 600D form the resonant circuit together with the path which goes around the opening 512, the additional opening 516, the slit 514D and the additional slit 518 of the conductive portion 510D. A resonant frequency of the resonant circuit is a communication frequency band of the antenna device 10D. The capacitors 600D are mounted on the circuit board 500D so that each of the capacitors 600D is wired across the additional slit 518. However, the mounting locations of the capacitors 600D are not limited to those described above, but the capacitors 600D may be mounted at any locations on the conductive portion 510D, provided that the capacitors 600D and the path, which goes around the opening 512, the additional opening 516, the slit 514D and the additional slit 518 of the conductive portion 510D, form the current loop path.

Fourth Embodiment

[0132] As shown in FIG. 29, an antenna device 10E according to a fourth embodiment of the present invention comprises a circuit board 500E, a coil antenna 100 and a resonance circuit. The coil antenna 100 of the present embodiment has a structure same as that of the coil antenna 100 of the first embodiment, and detail explanation thereabout will be omitted. As for directions and orientations in the present embodiment, expressions same as those of the first embodiment will be used hereinbelow.

[0133] As shown in FIG. 29, the circuit board 500E of the present embodiment has a conductive portion 510E with a flat shape. In more detail, the conductive portion 510E is a ground conductor. Specifically, the circuit board 500E is configured so that a conductive layer, which functions as the conductive portion 510E, is laminated on an insulating substrate such as a glass epoxy substrate.

[0134] As shown in FIG. 29, the conductive portion 510E is formed with an opening 512 and a slit 514E. The conductive portion 510E has a first path which goes around the opening 512 and the slit 514E. The first path forms a part of the resonance circuit. The opening 512 of the present embodiment has a structure same as that of the opening 512 of the first embodiment, and detail explanation thereabout will be omitted.

[0135] As shown in FIG. 29, similar to the opening 512, the slit 514E of the present embodiment is a part of the circuit board 500E which has no conductive pattern. The slit 514E is coupled with the opening 512. The slit 514E extends leftward in the right-left direction from the opening 512, and then extends rearward in the front-rear direction. The slit

514E extends from the opening 512. The slit 514E does not communicate with the outside of the conductive portion 510E in a plane perpendicular to the up-down direction.

[0136] As shown in FIG. 29, a width of the slit 514E is smaller than a diameter of the opening 512. However, the present invention is not limited thereto, but the width of the slit 514E may be equal to the diameter of the opening 512, or the width of the slit 514E may be greater than the diameter of the opening 512. It is noted that an increase of the width of the slit 514E enlarges a current loop, which is formed around the slit 514E, so that a communication range of the antenna device 10E is increased.

[0137] As shown in FIG. 29, the conductive portion 510E is further formed with an additional opening 516.

[0138] As shown in FIG. 29, the additional opening 516 of the present embodiment is a hole piercing the circuit board 500E. In other words, the additional opening 516 pierces the circuit board 500E in the up-down direction. However, the present invention is not limited thereto, but the additional opening 516 may be a part of the circuit board 500E, which has no conductive pattern, similar to the opening 512. The additional opening 516 is positioned rearward of the opening 512 in the front-rear direction. In the plane perpendicular to the up-down direction, the additional opening 516 is greater than the opening 512. The conductive portion 510E has a second path which goes around the additional opening 516. The second path forms a part of the resonance circuit. The slit 514E is coupled with both of the opening 512 and the additional opening 516.

[0139] As shown in FIG. 29, the conductive portion 510E is further formed with an additional slit 518.

[0140] As shown in FIG. 29, the additional slit 518 of the present embodiment is a part of the circuit board 500E, which has no conductive pattern, similar to the opening 512. The additional slit 518 is coupled with the additional opening 516. The additional slit 518 extends from the additional opening 516. The additional slit 518 extends in the right-left direction. The additional slit 518 communicates with the outside of the conductive portion 510E in the plane perpendicular to the up-down direction. All of the opening 512, the slit 514E, the additional opening 516 and the additional slit 518 are coupled to one another.

[0141] As shown in FIG. 29, the circuit board 500E has an edge 520.

[0142] As shown in FIG. 29, the edge 520 defines an outer edge of the circuit board 500E. Specifically, the edge 520 defines the outer edge of the circuit board 500E in a direction perpendicular to the up-down direction. The slit 514E does not reach the edge 520. The additional slit 518 extends toward the edge 520. Specifically, the additional slit 518 extends from the additional opening 516 toward the edge 520. The additional slit 518 reaches the edge 520.

[0143] Referring to FIG. 29, the coil antenna 100 is mounted on the circuit board 500E so that the coil antenna 100 at least overlaps with the opening 512 when the coil antenna 100 is viewed along the up-down direction perpendicular to the circuit board 500E. The coil antenna 100 is mounted on the circuit board 500E so that the coil antenna 100 does not overlap with the additional opening 516 when the coil antenna 100 is viewed along the up-down direction. The coil antenna 100 is surface-mounted on the circuit board 500E.

[0144] As shown in FIG. 29, the antenna device 10E further has capacitors 600E which form a part of the resonant circuit.

[0145] Referring to FIG. 29, the capacitors 600E form a current loop path together with a path which goes around the opening 512, the additional opening 516, the slit 514E and the additional slit 518 of the conductive portion 510E. In other words, the capacitors 600E form the resonant circuit together with the path which goes around the opening 512, the additional opening 516, the slit 514E and the additional slit 518 of the conductive portion 510E. A resonant frequency of the resonant circuit is a communication frequency band of the antenna device 10E. The capacitors 600E are mounted on the circuit board 500E so that each of the capacitors 600E is wired across the additional slit 518. However, the mounting locations of the capacitors 600E are not limited to those described above, but the capacitors 600E may be mounted at any locations on the conductive portion 510E, provided that the capacitors 600E and the path, which goes around the opening 512, the additional opening 516, the slit 514E and the additional slit 518 of the conductive portion 510E, form the current loop path.

Fifth Embodiment

[0146] As shown in FIG. 30, an antenna device 10F according to a fifth embodiment of the present invention comprises a circuit board 500F, a coil antenna 100 and a resonance circuit. The coil antenna 100 of the present embodiment has a structure same as that of the coil antenna 100 of the first embodiment, and detail explanation thereabout will be omitted. As for directions and orientations in the present embodiment, expressions same as those of the first embodiment will be used hereinbelow.

[0147] As shown in FIG. 30, the circuit board 500F of the present embodiment has a conductive portion 510F with a flat shape. In more detail, the conductive portion 510F is a ground conductor. Specifically, the circuit board 500F is configured so that a conductive layer, which functions as the conductive portion 510F, is laminated on an insulating substrate such as a glass epoxy substrate.

[0148] As shown in FIG. 30, the conductive portion 510F is formed with an opening 512 and a slit 514F. The conductive portion 510F has a first path which goes around the opening 512 and the slit 514F. The first path forms a part of the resonance circuit. The opening 512 of the present embodiment has a structure same as that of the opening 512 of the first embodiment, and detail explanation thereabout will be omitted.

[0149] As shown in FIG. 30, similar to the opening 512, the slit 514F of the present embodiment is a part of the circuit board 500F which has no conductive pattern. The slit 514F is coupled with the opening 512. The slit 514F extends from the opening 512.

[0150] As shown in FIG. 30, a width of the slit 514F is smaller than a diameter of the opening 512. However, the present invention is not limited thereto, but the width of the slit 514F may be equal to the diameter of the opening 512, or the width of the slit 514F may be greater than the diameter of the opening 512. It is noted that an increase of the width of the slit 514F enlarges a current loop, which is formed around the slit 514F, so that a communication range of the antenna device 10F is increased.

[0151] As shown in FIG. 30, the conductive portion 510F is further formed with an additional opening 516.

[0152] As shown in FIG. 30, the additional opening 516 of the present embodiment is a hole piercing the circuit board 500F. In other words, the additional opening 516 pierces the circuit board 500F in the up-down direction. However, the present invention is not limited thereto, but the additional opening 516 may be a part of the circuit board 500F, which has no conductive pattern, similar to the opening 512. The additional opening 516 is positioned rearward of the opening 512 in the front-rear direction. In a plane perpendicular to the up-down direction, the additional opening 516 is greater than the opening 512. The conductive portion 510F has a second path which goes around the additional opening 516. The second path forms a part of the resonance circuit. The slit 514F is coupled with both of the opening 512 and the additional opening 516.

[0153] As shown in FIG. 30, the conductive portion 510F is further formed with an additional slit 518.

[0154] As shown in FIG. 30, the additional slit 518 of the present embodiment is a part of the circuit board 500F, which has no conductive pattern, similar to the opening 512. The additional slit 518 is coupled with the slit 514F. The additional slit 518 extends in the right-left direction from the slit 514F. The additional slit 518 communicates with the outside of the conductive portion 510F in the plane perpendicular to the up-down direction. In the plane perpendicular to the up-down direction, the slit 514F communicates with the outside of the conductive portion 510F via the additional slit 518. All of the opening 512, the slit 514F, the additional opening 516 and the additional slit 518 are coupled to one another.

[0155] As shown in FIG. 30, a width of the additional slit 518 is equal to the width of the slit 514F. However, the present invention is not limited thereto, but the width of the additional slit 518 may be smaller than the width of the slit 514F or may be greater than the width of the slit 514F.

[0156] As shown in FIG. 30, the circuit board 500F has an edge 520.

[0157] As shown in FIG. 30, the edge 520 defines an outer edge of the circuit board 500F. Specifically, the edge 520 defines the outer edge of the circuit board 500F in a direction perpendicular to the up-down direction. The additional slit 518 extends toward the edge 520. Specifically, the additional slit 518 extends from the slit 514F toward the edge 520. The additional slit 518 reaches the edge 520.

[0158] Referring to FIG. 30, the coil antenna 100 is mounted on the circuit board 500F so that the coil antenna 100 at least overlaps with the opening 512 when the coil antenna 100 is viewed along the up-down direction perpendicular to the circuit board 500F. The coil antenna 100 is mounted on the circuit board 500F so that the coil antenna 100 does not overlap with the additional opening 516 when the coil antenna 100 is viewed along the up-down direction. The coil antenna 100 is surface-mounted on the circuit board 500F.

[0159] As shown in FIG. 30, the antenna device 10F further has capacitors 600F which form a part of the resonant circuit.

[0160] Referring to FIG. 30, the capacitors 600F form a current loop path together with a path which goes around the opening 512, the additional opening 516, the slit 514F and the additional slit 518 of the conductive portion 510F. In other words, the capacitors 600F form the resonant circuit together with the path which goes around the opening 512, the additional opening 516, the slit 514F and the additional

slit **518** of the conductive portion **510F**. A resonant frequency of the resonant circuit is a communication frequency band of the antenna device **10F**. The capacitors **600F** are mounted on the circuit board **500F** so that each of the capacitors **600F** is wired across the additional slit **518**. However, the mounting locations of the capacitors **600F** are not limited to those described above, but the capacitors **600F** may be mounted at any locations on the conductive portion **510F**, provided that the capacitors **600F** and the path, which goes around the opening **512**, the additional opening **516**, the slit **514F** and the additional slit **518** of the conductive portion **510F**, form the current loop path.

[0161] Although the specific explanation about the present invention is made above referring to the embodiments, the present invention is not limited thereto and is susceptible to various modifications and alternative forms. In addition, the above embodiments and variations may also be combined.

[0162] Although each of the antenna devices **10**, **10C** of the aforementioned first and second embodiments is not provided with a portion equivalent to the additional opening **516** of each of the antenna devices **10D**, **10E**, **10F** of the third, fourth and fifth embodiments, the present invention is not limited thereto. Specifically, the antenna device **10**, **10C** may also be provided with an additional opening **516** which forms a part of the slit **514**, **514C**.

[0163] The present application is based on a Japanese patent applications of JP2022-074955 filed before the Japan Patent Office on Apr. 28, 2022 and JP2022-118580 filed before the Japan Patent Office on Jul. 26, 2022, the contents of which are incorporated herein by reference.

[0164] While there has been described what is believed to be the preferred embodiment of the invention, those skilled in the art will recognize that other and further modifications may be made thereto without departing from the spirit of the invention, and it is intended to claim all such embodiments that fall within the true scope of the invention.

REFERENCE SIGNS LIST

[0165] **10**, **10C**, **10D**, **10E**, **10F** antenna device
 [0166] **100**, **100A**, **100B** coil antenna
 [0167] **200** coil
 [0168] **210** upper end
 [0169] **220** lower end
 [0170] **230** outer periphery
 [0171] **240** inner periphery
 [0172] **300**, **300A**, **300B** magnetic core
 [0173] **310** center core
 [0174] **320**, **320B** upper portion
 [0175] **340** side portion
 [0176] **400** terminal
 [0177] **500**, **500C**, **500D**, **500E**, **500F** circuit board
 [0178] **510**, **510C**, **510D**, **510E**, **510F** conductive portion
 [0179] **512** opening
 [0180] **514**, **514C**, **514D**, **514E**, **514F** slit
 [0181] **5142** first portion
 [0182] **5144** second portion
 [0183] **5146** third portion
 [0184] **516** additional opening
 [0185] **518** additional slit
 [0186] **520** edge
 [0187] **600**, **600D**, **600E**, **600F** capacitor
 [0188] **700**, **700A** molded resinous member
 [0189] **AX** winding axis

[0190] **PP** predetermined portion

[0191] **SR** specific region

1. An antenna device comprising a circuit board, a coil antenna and a resonance circuit, wherein:

the circuit board has a conductive portion with a flat shape;

the conductive portion is formed with an opening and a slit;

the conductive portion has a first path which goes around the opening and the slit;

the first path forms a part of the resonance circuit;

the slit is coupled with the opening;

the coil antenna is mounted on the circuit board so that the coil antenna at least overlaps with the opening when the coil antenna is viewed along an up-down direction perpendicular to the circuit board;

the coil antenna comprises a coil and a magnetic core;

the coil has a winding axis extending in the up-down direction;

the magnetic core partially forms a magnetic path of the coil;

the magnetic core is positioned at a region excluding a specific region which is positioned just below the coil;

the magnetic core at least has a center core; and

the center core is at least partially positioned inside the coil.

2. The antenna device as recited in claim **1**, wherein:

the circuit board has an edge; and

the slit extends toward the edge.

3. The antenna device as recited in claim **1**, wherein:

the conductive portion is further formed with an additional opening;

the conductive portion has a second path which goes around the additional opening;

the second path forms a part of the resonance circuit;

the slit is coupled with both of the opening and the additional opening; and

the coil antenna is mounted on the circuit board so that the coil antenna does not overlap with the additional opening when the coil antenna is viewed along the up-down direction.

4. The antenna device as recited in claim **3**, wherein:

the circuit board has an edge;

the conductive portion is further formed with an additional slit;

the additional slit extends toward the edge;

the antenna device further has a capacitor which forms a part of the resonant circuit; and

the capacitor is mounted on the circuit board so that the capacitor is wired across the additional slit.

5. The antenna device as recited in claim **4**, wherein all of the opening, the slit, the additional opening and the additional slit are coupled to one another.

6. The antenna device as recited in claim **1**, wherein:

the coil has a lower end in the up-down direction; and

the center core extends downward in the up-down direction beyond the lower end.

7. The antenna device as recited in claim **1**, wherein:

the coil has an upper end in the up-down direction; and the center core extends upward in the up-down direction beyond the upper end.

8. The antenna device as recited in claim **7**, wherein:

the coil has an outer periphery;

the magnetic core further has an upper portion;

the upper portion extends from the center core toward the outer periphery; and

the upper portion overlaps with the coil when the upper portion is viewed in the up-down direction.

9. The antenna device as recited in claim 8, wherein: the magnetic core further has a side portion; the side portion is positioned outside the coil in a plane perpendicular to the up-down direction; and the side portion extends downward in the up-down direction from the upper portion.

10. The antenna device as recited in claim 1, wherein: the circuit board has an edge; the slit has a predetermined portion; and the predetermined portion extends along the edge in the vicinity of the edge.

11. The antenna device as recited in claim 10, wherein: the antenna device further has a capacitor which forms a part of the resonant circuit; and

the capacitor is mounted on the circuit board at its region between the edge and the slit.

12. The antenna device as recited in claim 10, wherein: the slit has a first portion and a second portion; the first portion extends from the opening toward the edge;

the second portion extends from the first portion in a direction intersecting with the first portion; and the second portion functions as the predetermined portion.

13. The antenna device as recited in claim 12, wherein: the slit further comprises a third portion; and the third portion extends from the second portion in a direction intersecting with the second portion.

14. The antenna device as recited in claim 1, wherein: the coil has an inner periphery; and $S1$ and $S2$ satisfy $1 \leq S2/S1 \leq 2$, where $S1$ is a cross-sectional area of the inner periphery of the coil, and $S2$ is an area of the opening.

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